

BFS Traversal

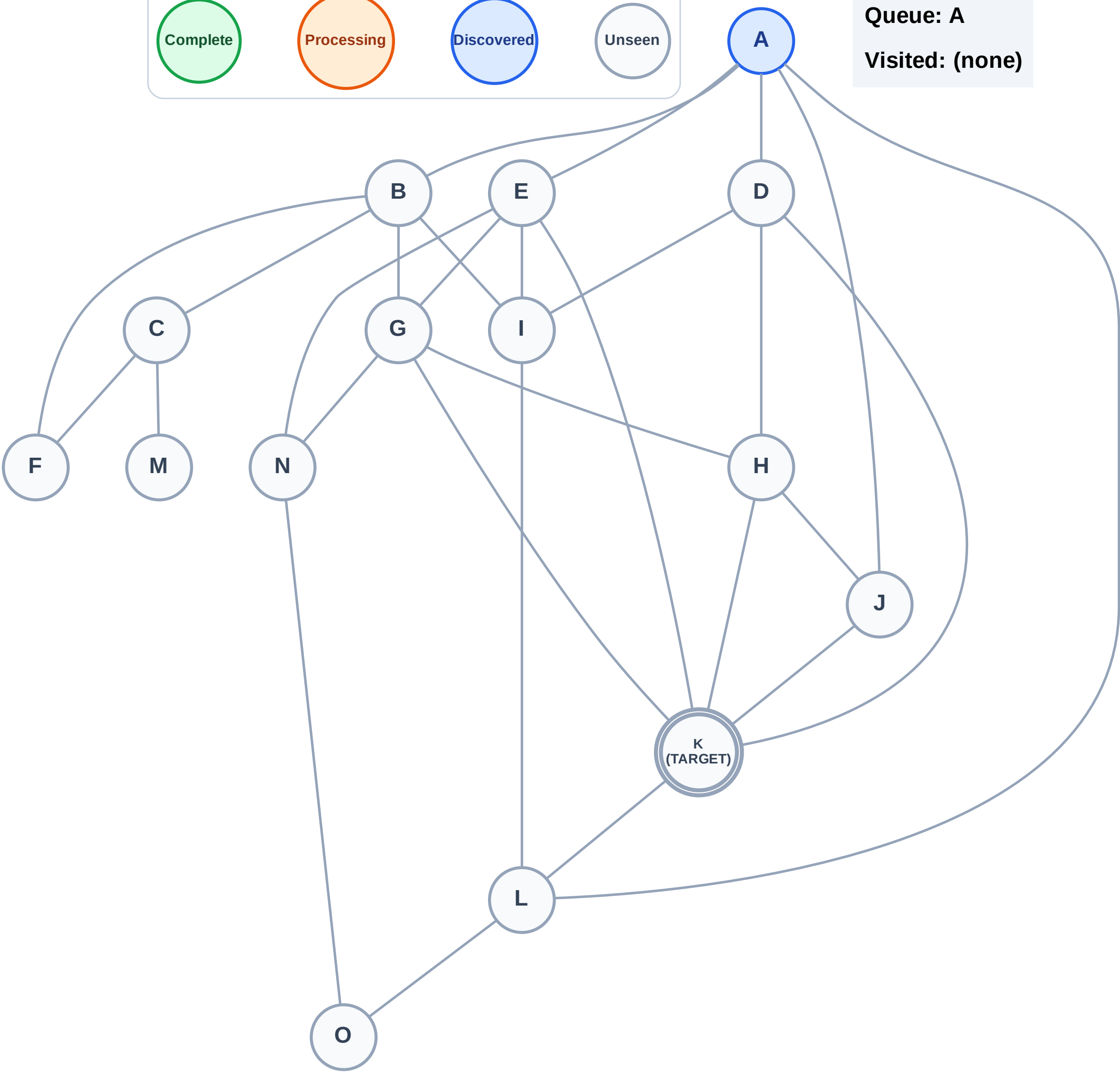
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



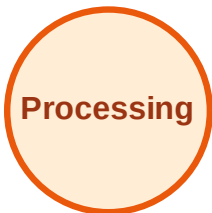
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



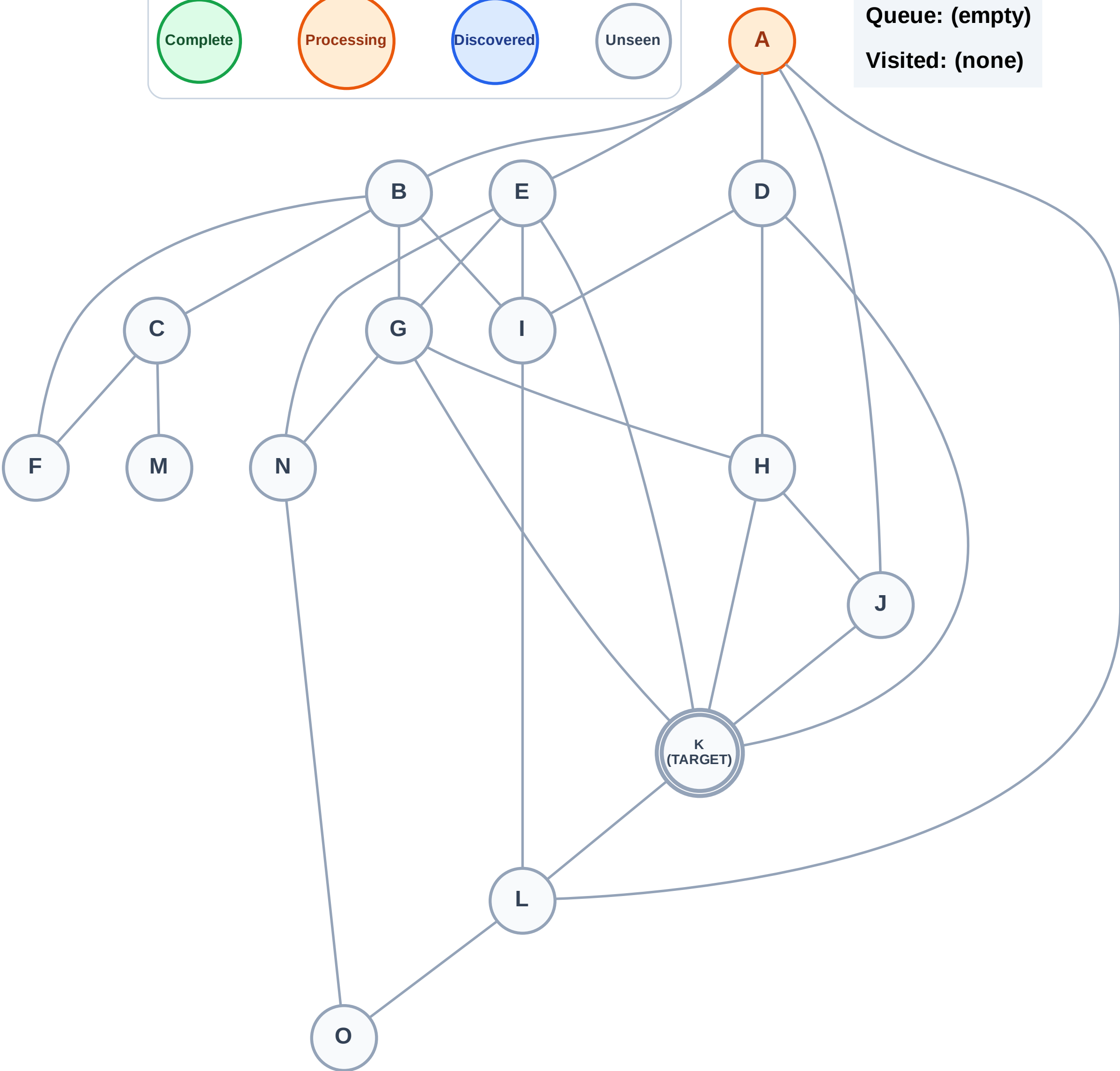
Discovered



Unseen

Queue: (empty)

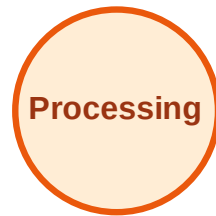
Visited: (none)



BFS Traversal

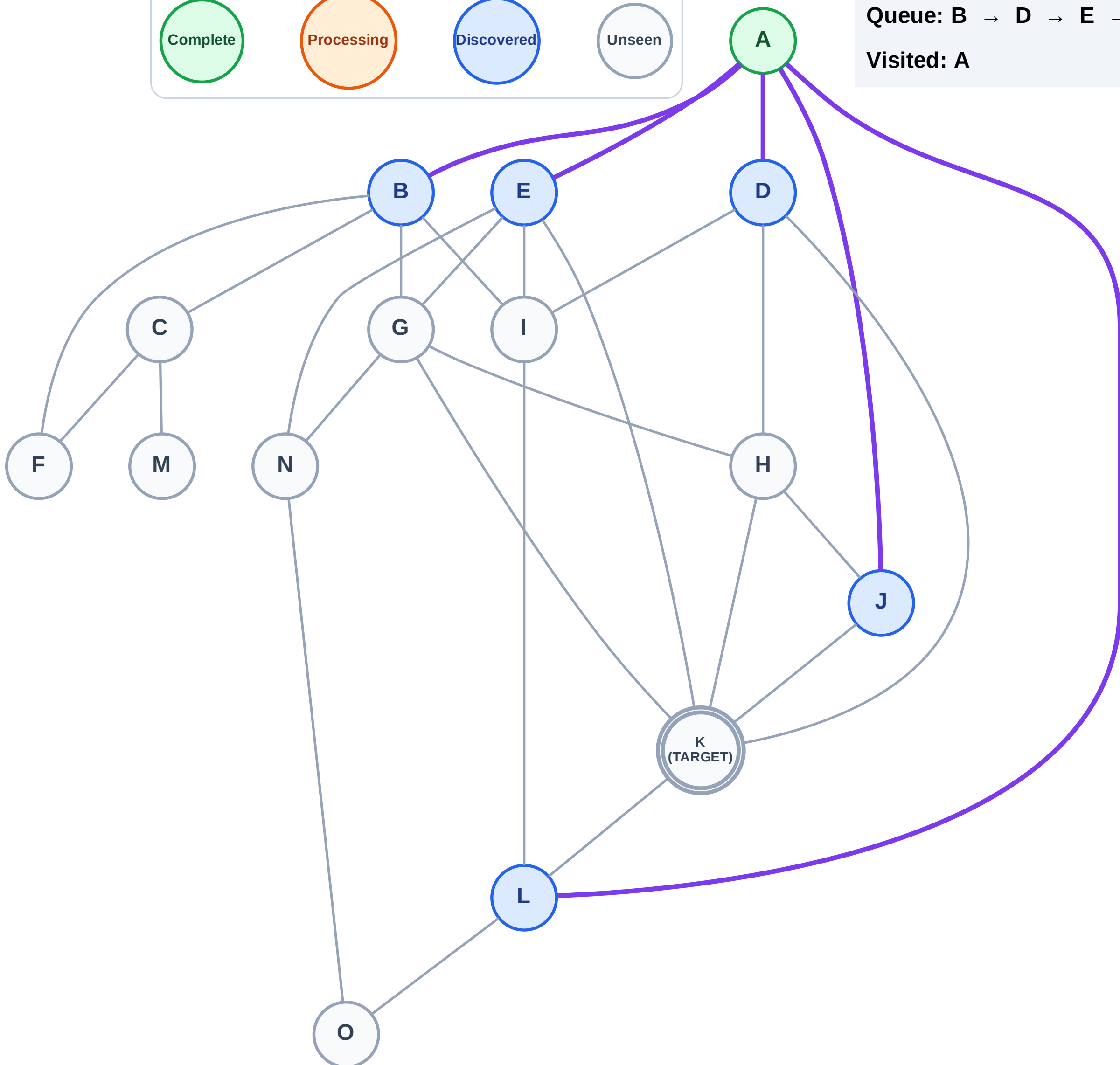
Step 3: Discover B, D, E, J, L from A

Legend



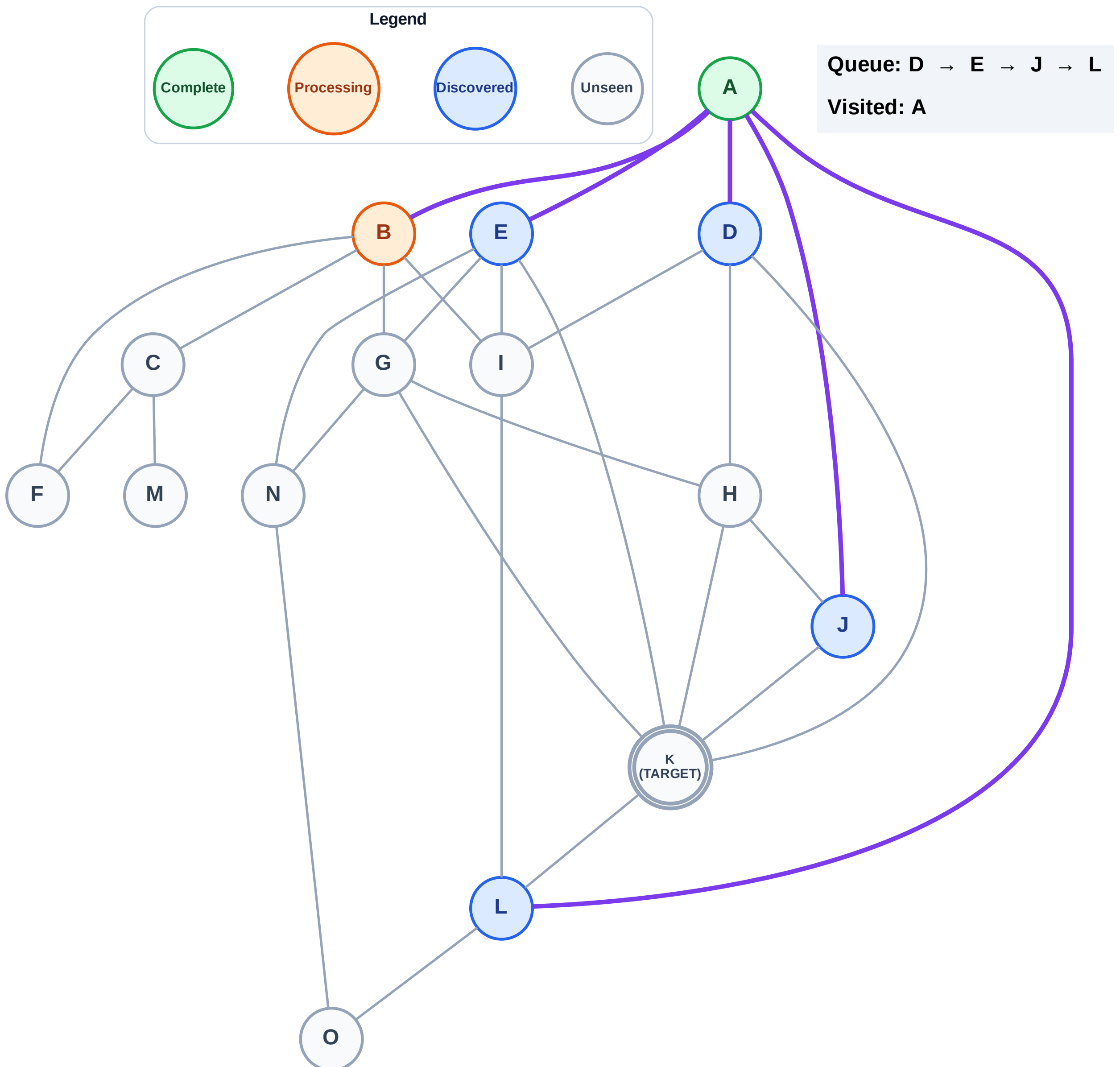
Queue: B → D → E → J → L

Visited: A

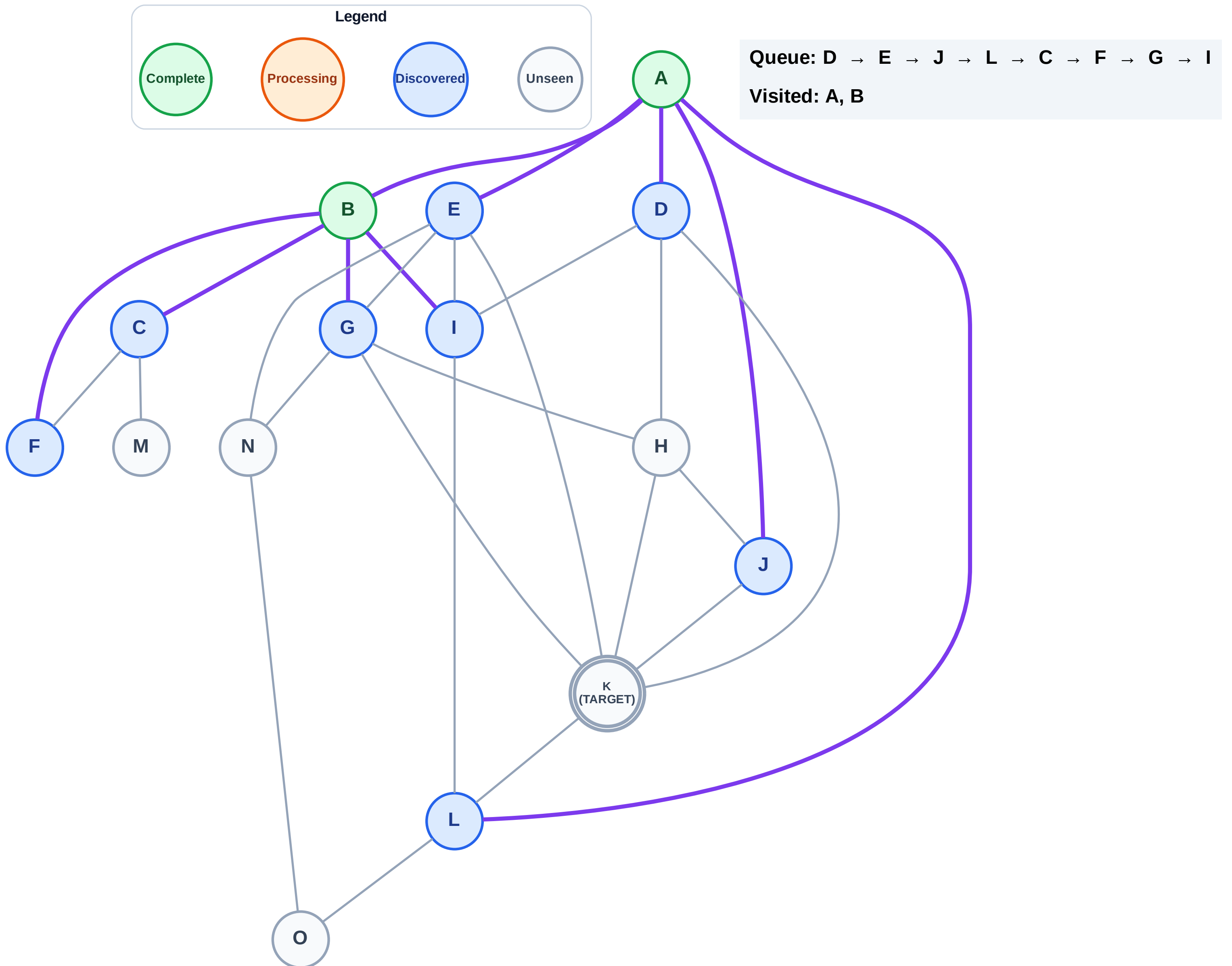


Step 4: Process node B

Step 4: Process node B



Step 5: Discover C, F, G, I from B



BFS Traversal

Step 6: Process node D

Legend

Complete

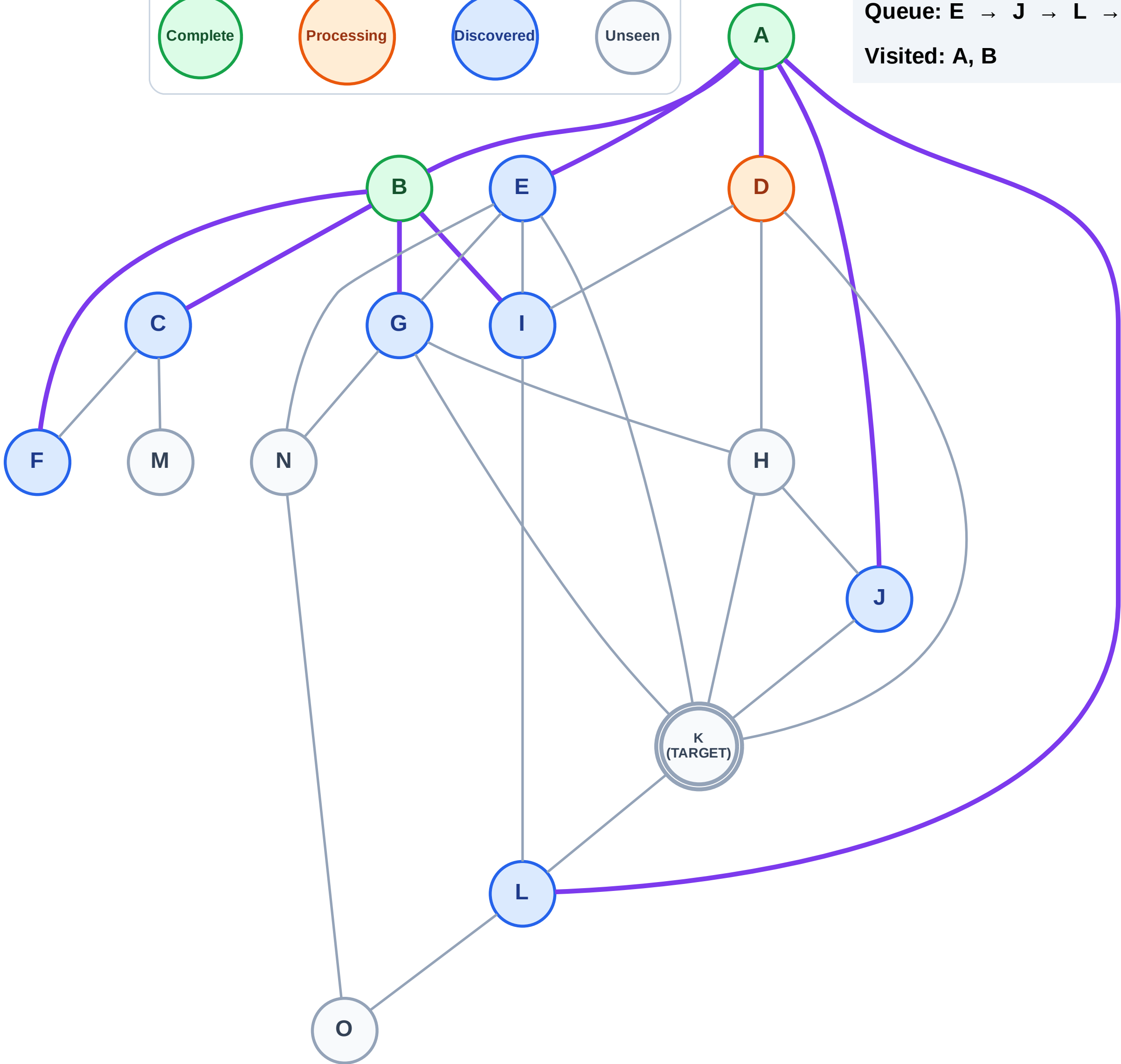
Processing

Discovered

Unseen

Queue: E → J → L → C → F → G → I

Visited: A, B



BFS Traversal

Step 7: Discover H, K from D

Legend

Complete

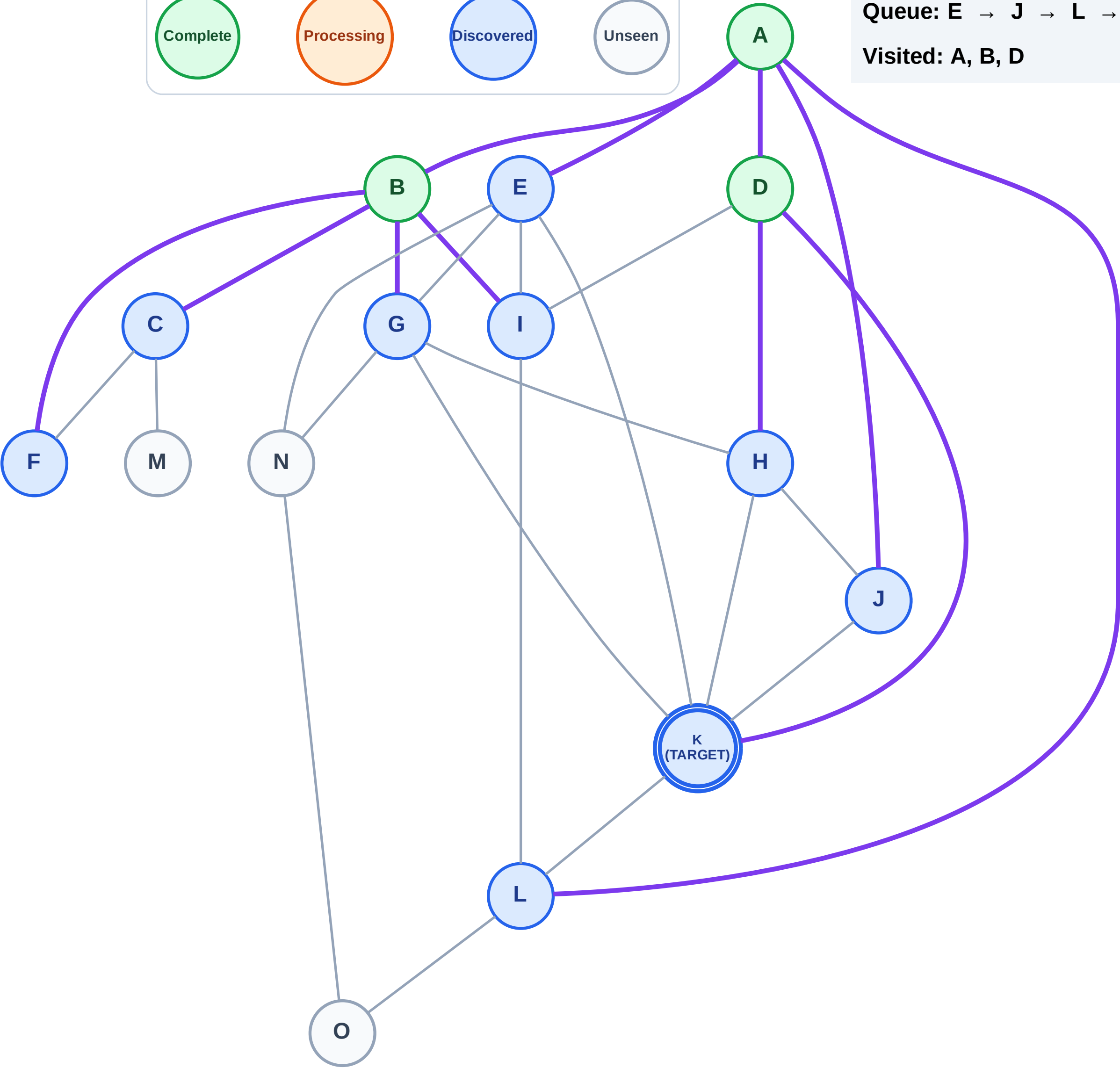
Processing

Discovered

Unseen

Queue: E → J → L → C → F → G → I → H → K

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

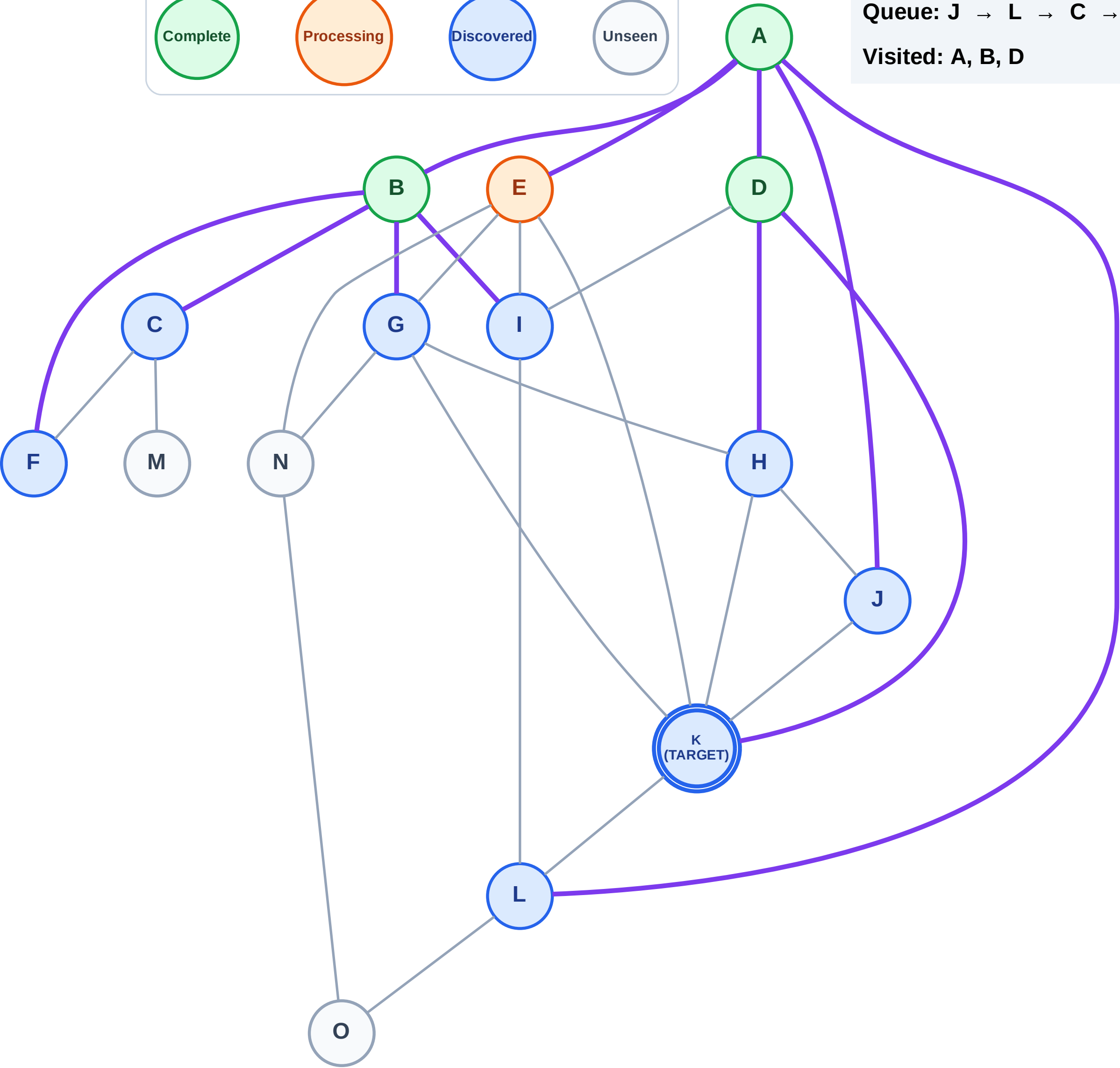
Processing

Discovered

Unseen

Queue: J → L → C → F → G → I → H → K

Visited: A, B, D



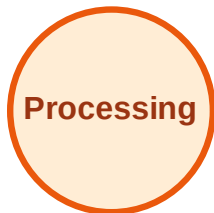
BFS Traversal

Step 9: Discover N from E

Legend



Complete



Processing



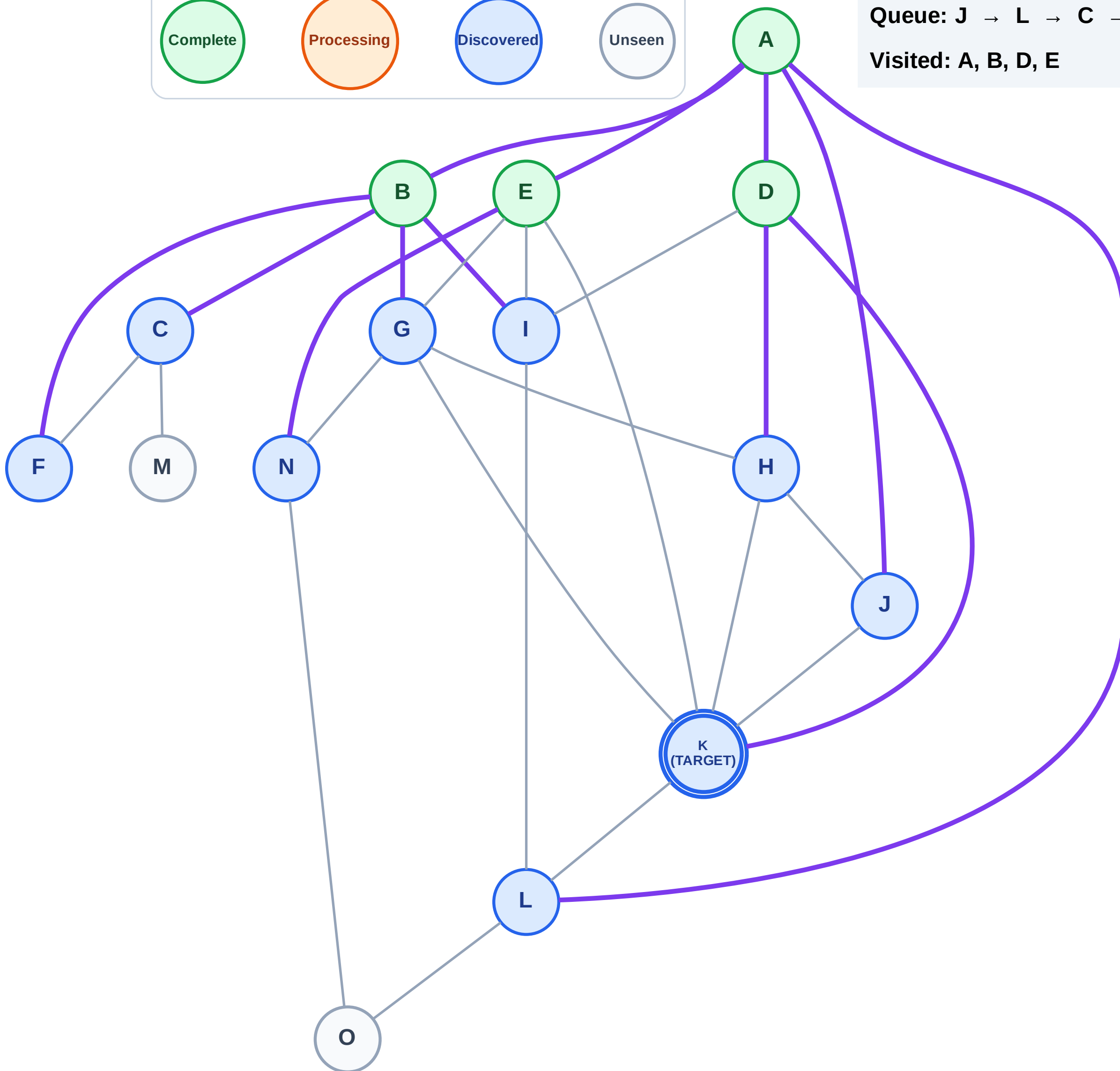
Discovered



Unseen

Queue: J → L → C → F → G → I → H → K → N

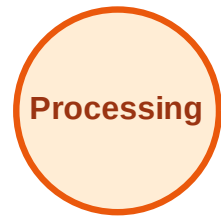
Visited: A, B, D, E



BFS Traversal

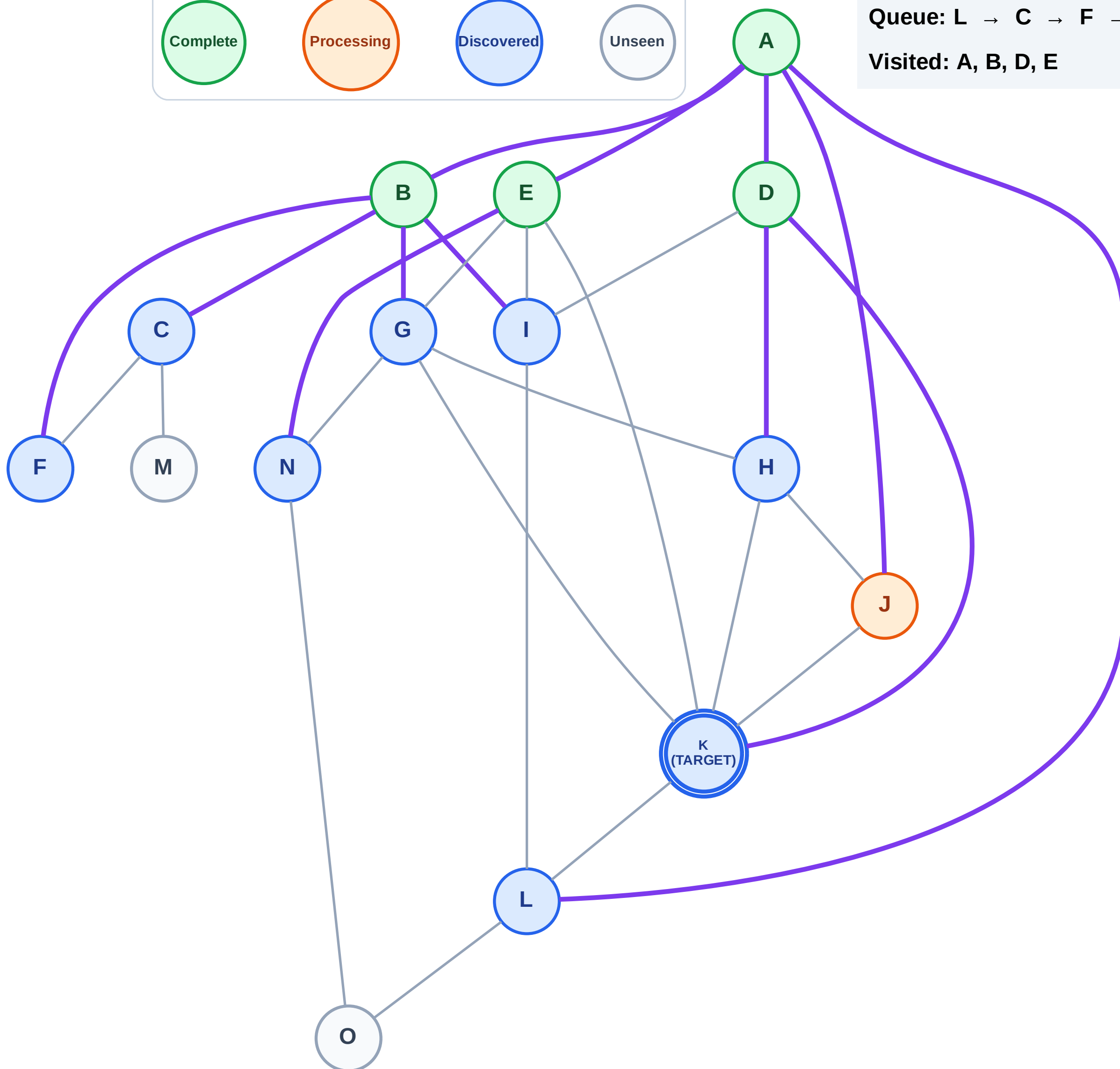
Step 10: Process node J

Legend



Queue: L → C → F → G → I → H → K → N

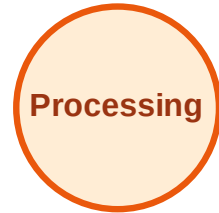
Visited: A, B, D, E



BFS Traversal

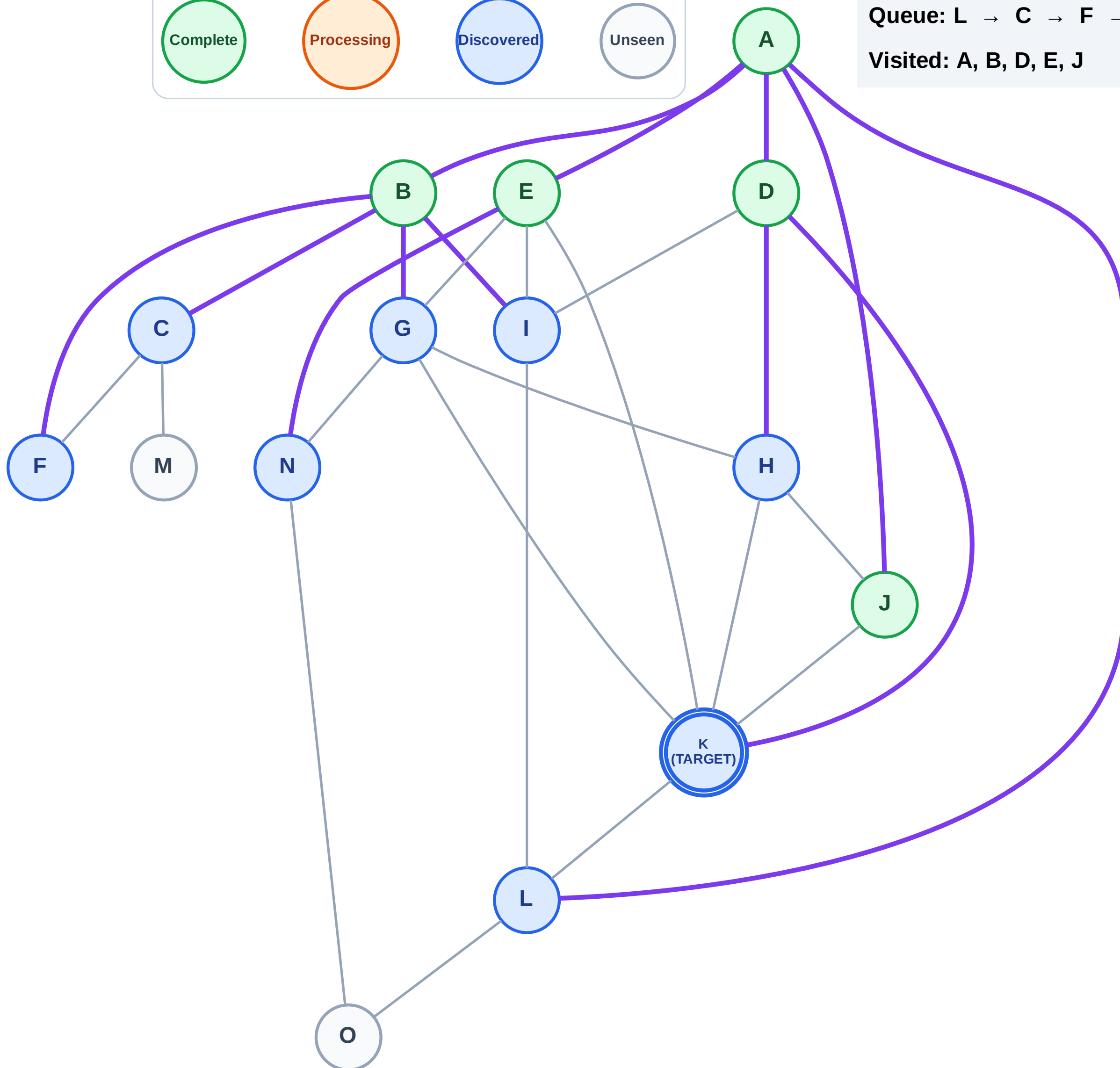
Step 11: No new neighbors from J

Legend



Queue: L → C → F → G → I → H → K → N

Visited: A, B, D, E, J



BFS Traversal

Step 12: Process node L

Legend

Complete

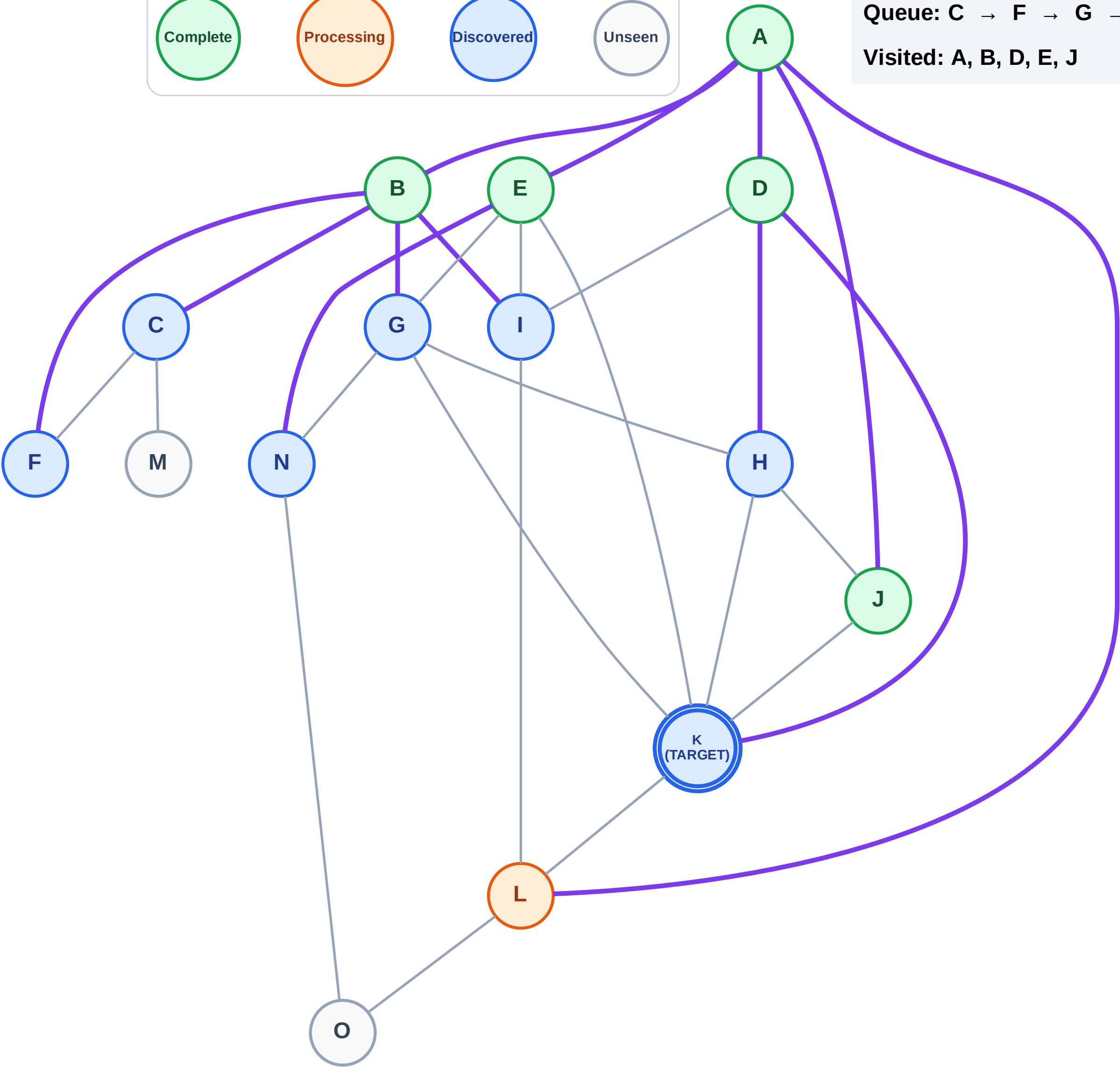
Processing

Discovered

Unseen

Queue: C → F → G → I → H → K → N

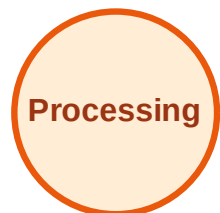
Visited: A, B, D, E, J



BFS Traversal

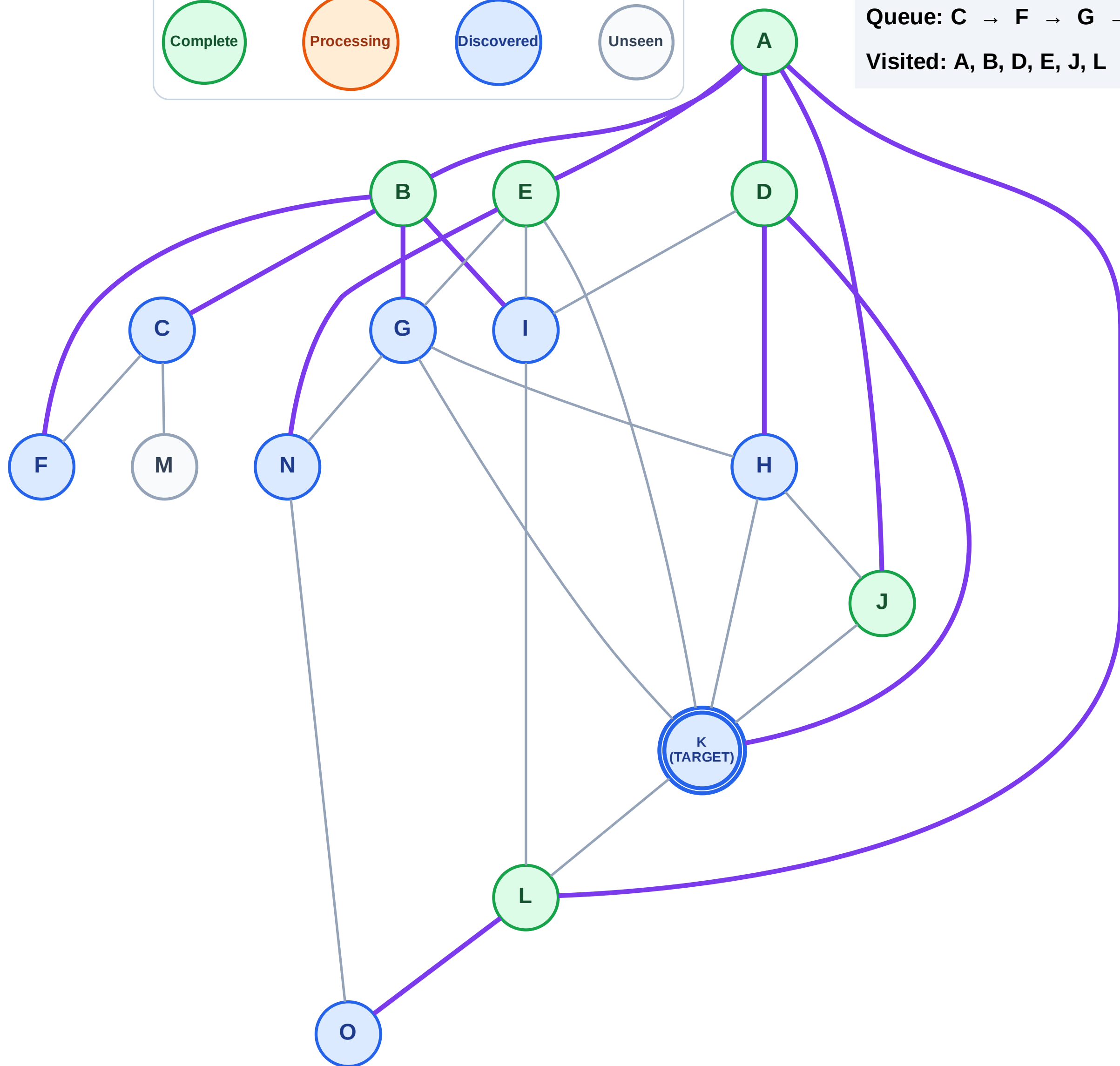
Step 13: Discover O from L

Legend



Queue: C → F → G → I → H → K → N → O

Visited: A, B, D, E, J, L



BFS Traversal

Step 14: Process node C

Legend

Complete

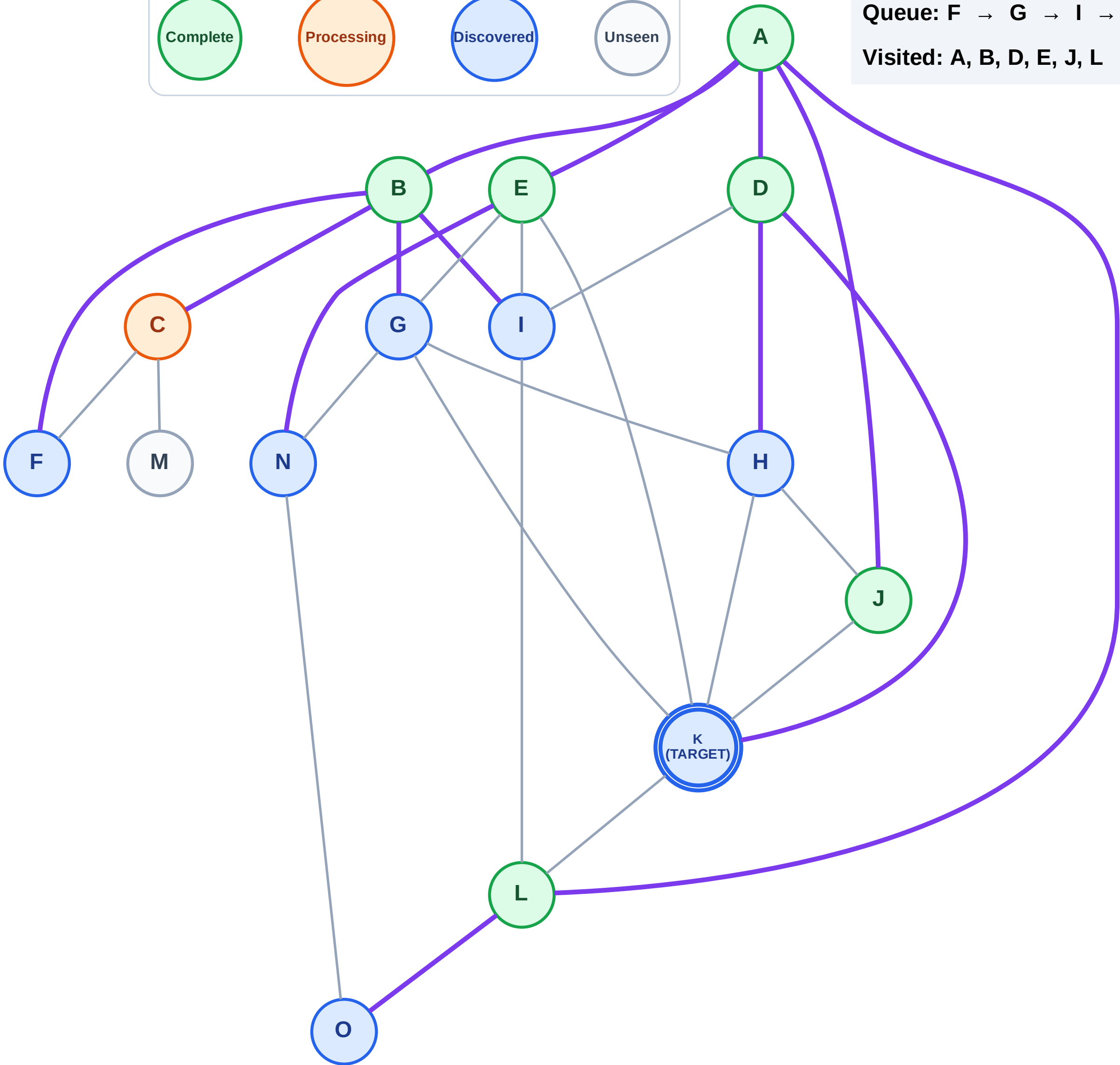
Processing

Discovered

Unseen

Queue: F → G → I → H → K → N → O

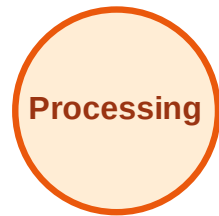
Visited: A, B, D, E, J, L



BFS Traversal

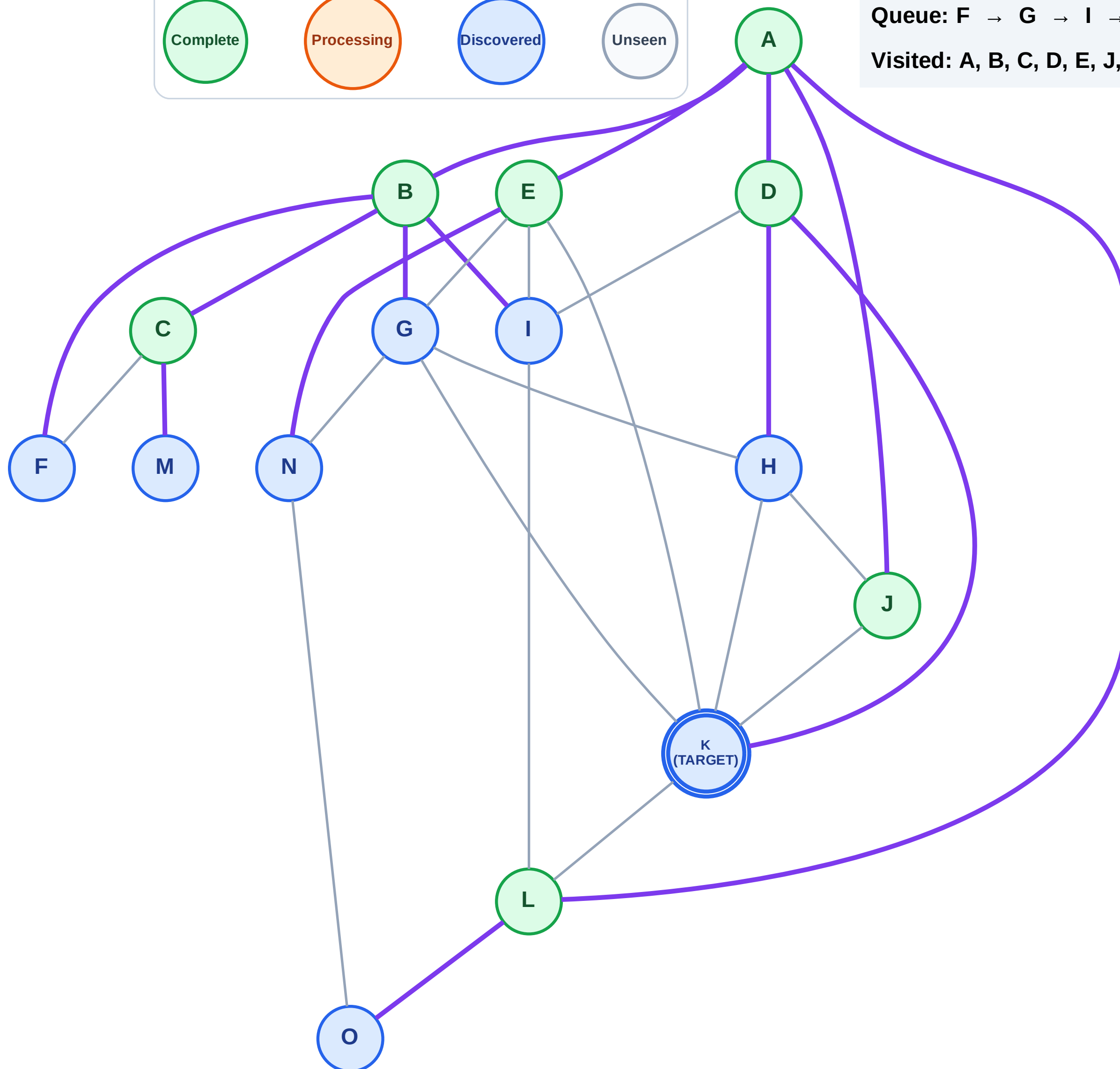
Step 15: Discover M from C

Legend



Queue: F → G → I → H → K → N → O → M

Visited: A, B, C, D, E, J, L



BFS Traversal

Step 16: Process node F

Legend

Complete

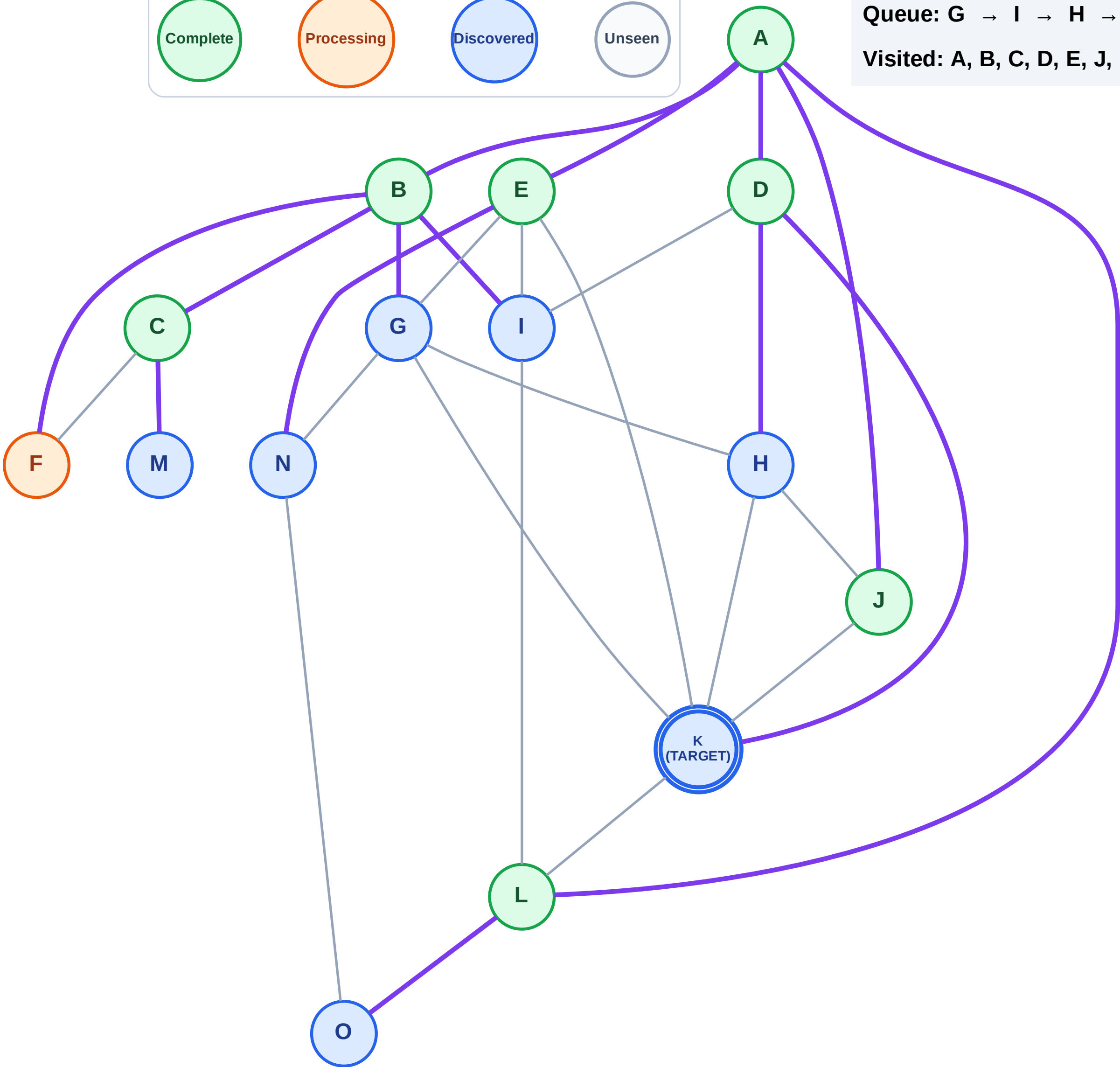
Processing

Discovered

Unseen

Queue: G → I → H → K → N → O → M

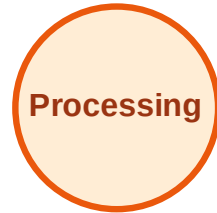
Visited: A, B, C, D, E, J, L



BFS Traversal

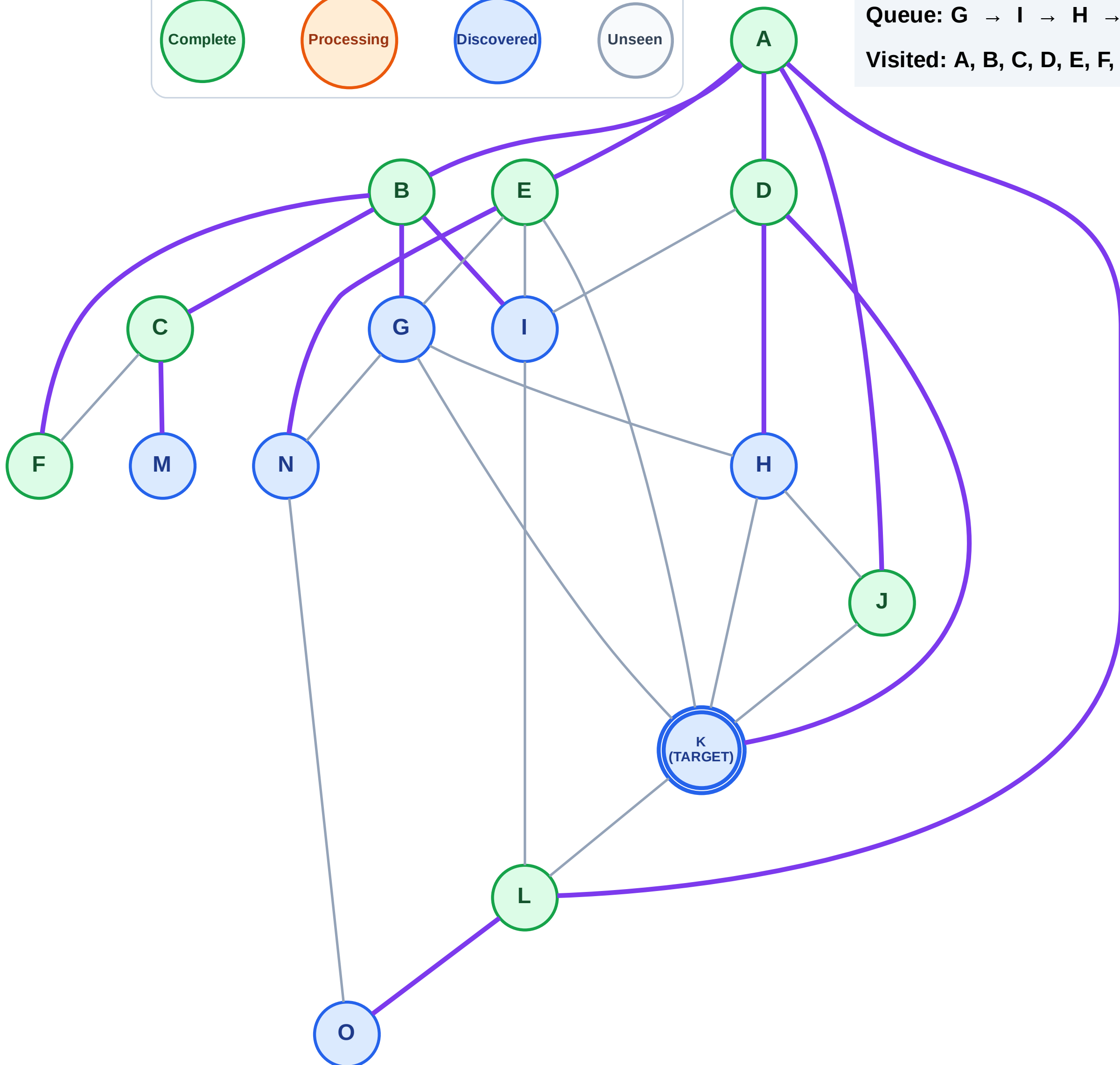
Step 17: No new neighbors from F

Legend



Queue: G → I → H → K → N → O → M

Visited: A, B, C, D, E, F, J, L



BFS Traversal

Step 18: Process node G

Legend

Complete

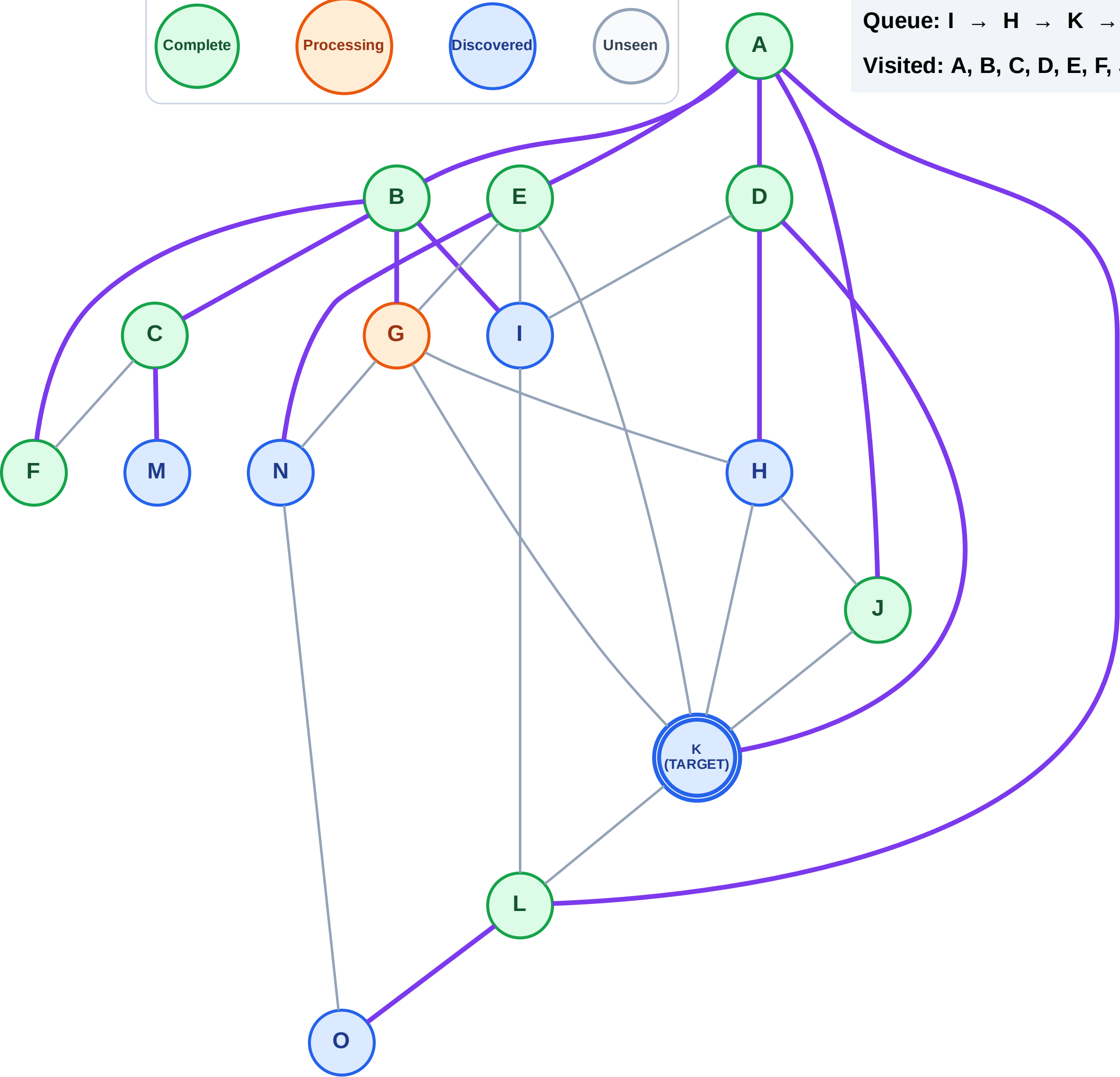
Processing

Discovered

Unseen

Queue: I → H → K → N → O → M

Visited: A, B, C, D, E, F, J, L



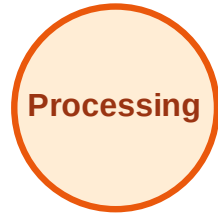
BFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



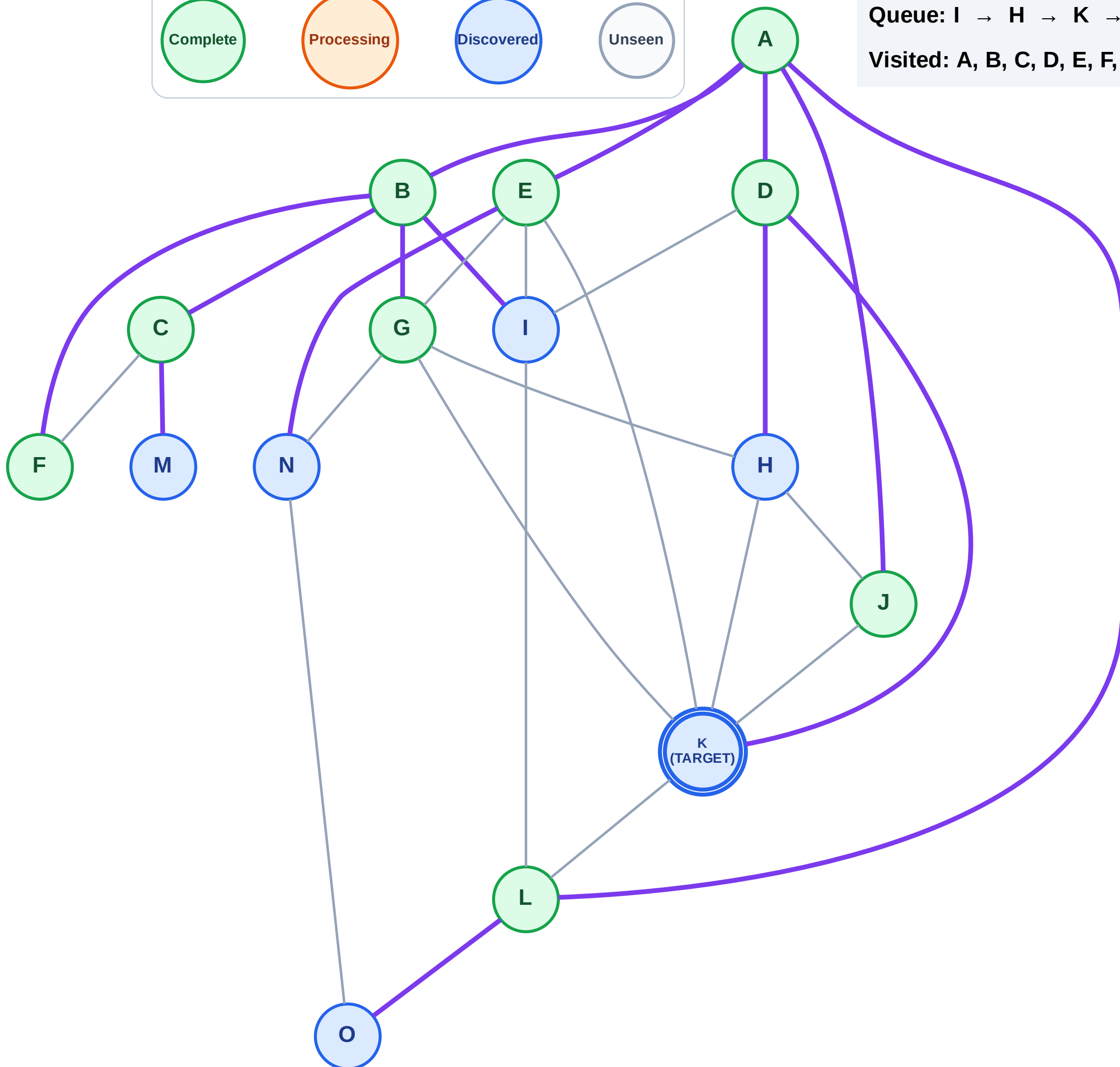
Discovered



Unseen

Queue: I → H → K → N → O → M

Visited: A, B, C, D, E, F, G, J, L



BFS Traversal

Step 20: Process node I

Legend

Complete

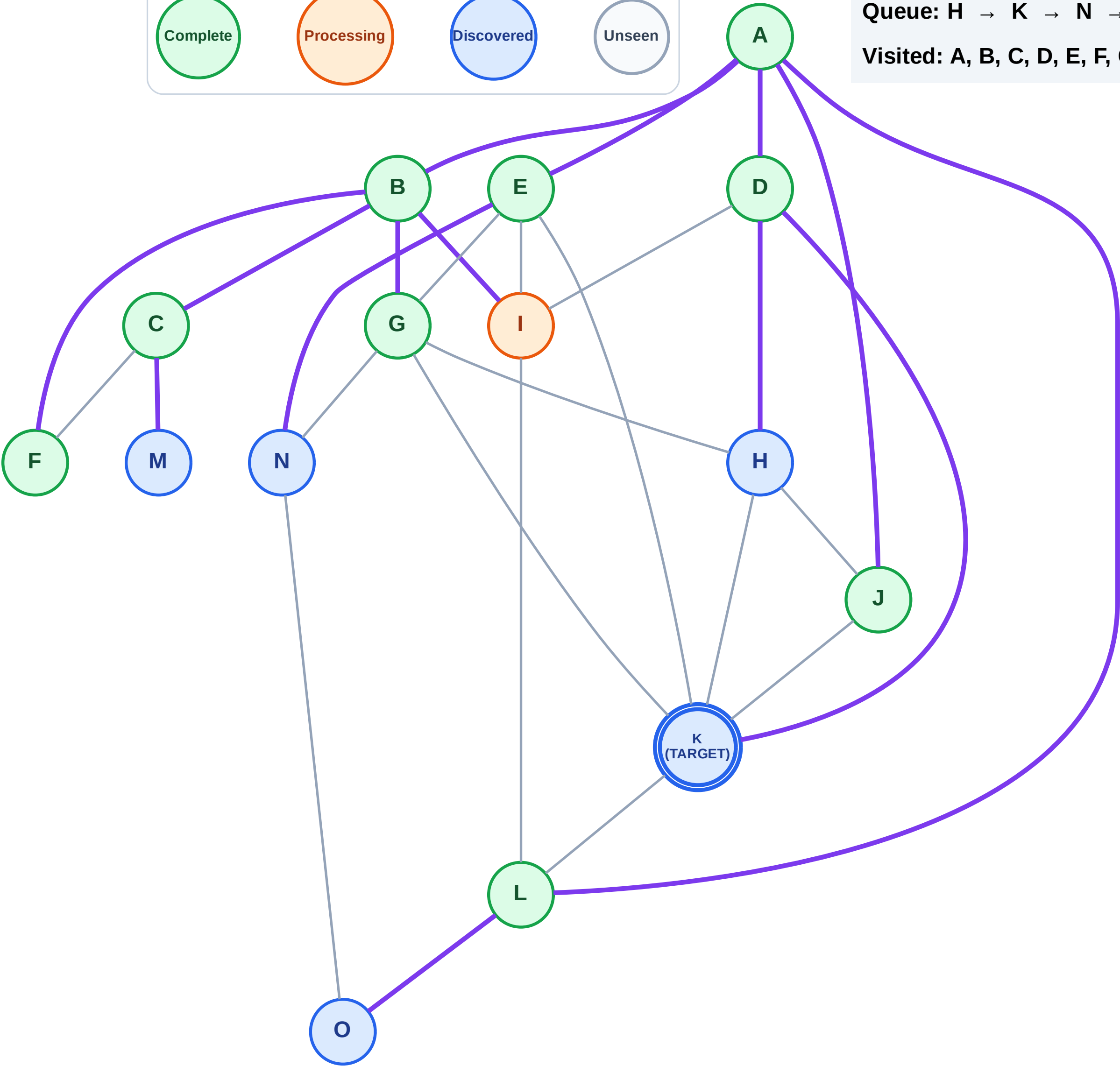
Processing

Discovered

Unseen

Queue: H → K → N → O → M

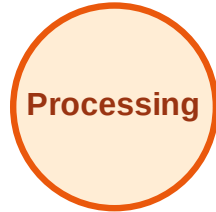
Visited: A, B, C, D, E, F, G, J, L



BFS Traversal

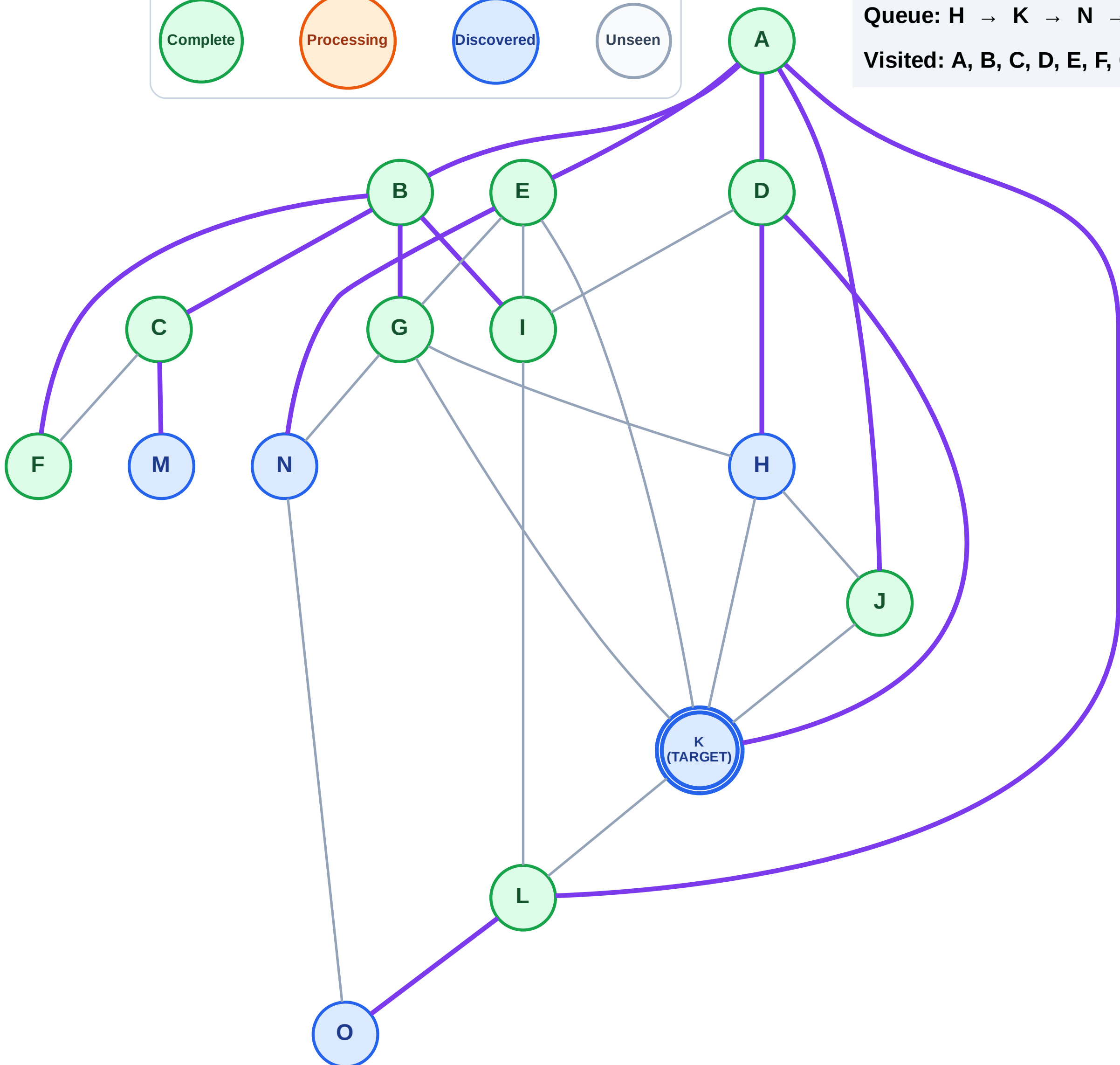
Step 21: No new neighbors from I

Legend



Queue: H → K → N → O → M

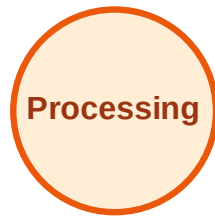
Visited: A, B, C, D, E, F, G, I, J, L



BFS Traversal

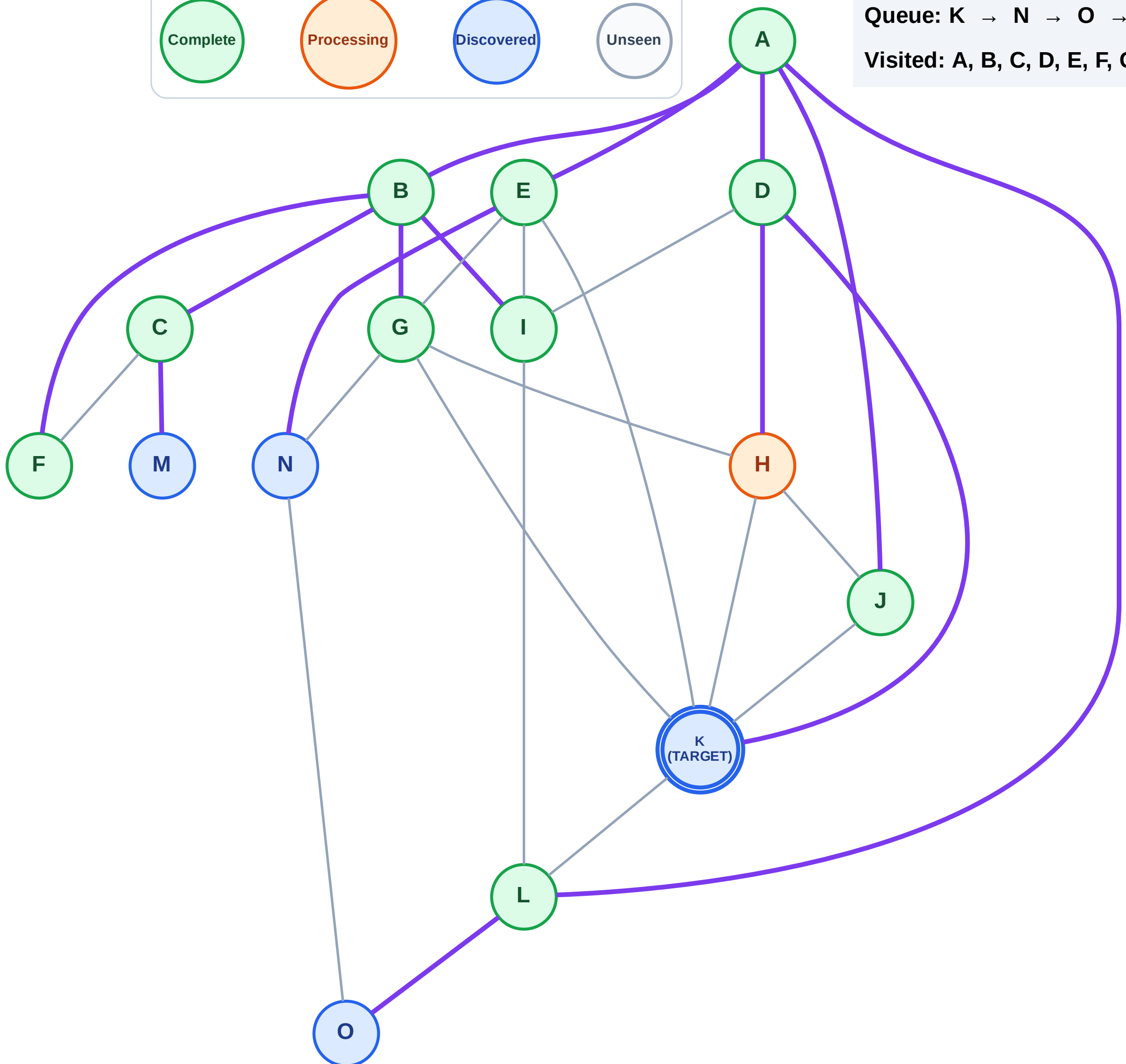
Step 22: Process node H

Legend



Queue: K → N → O → M

Visited: A, B, C, D, E, F, G, I, J, L



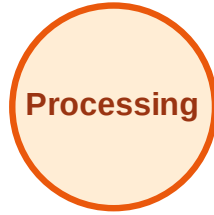
BFS Traversal

Step 23: No new neighbors from H

Legend



Complete



Processing



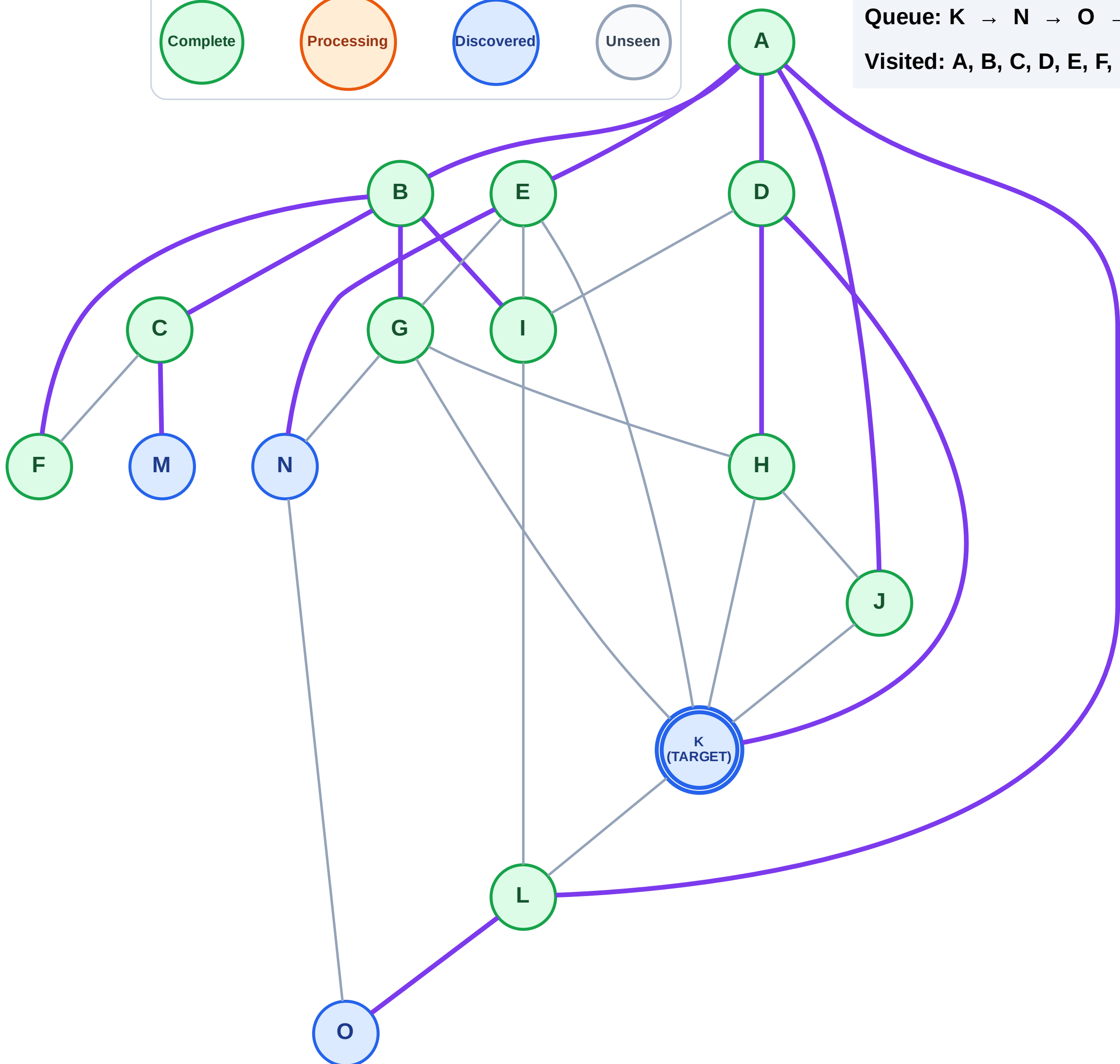
Discovered



Unseen

Queue: K → N → O → M

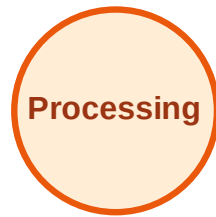
Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

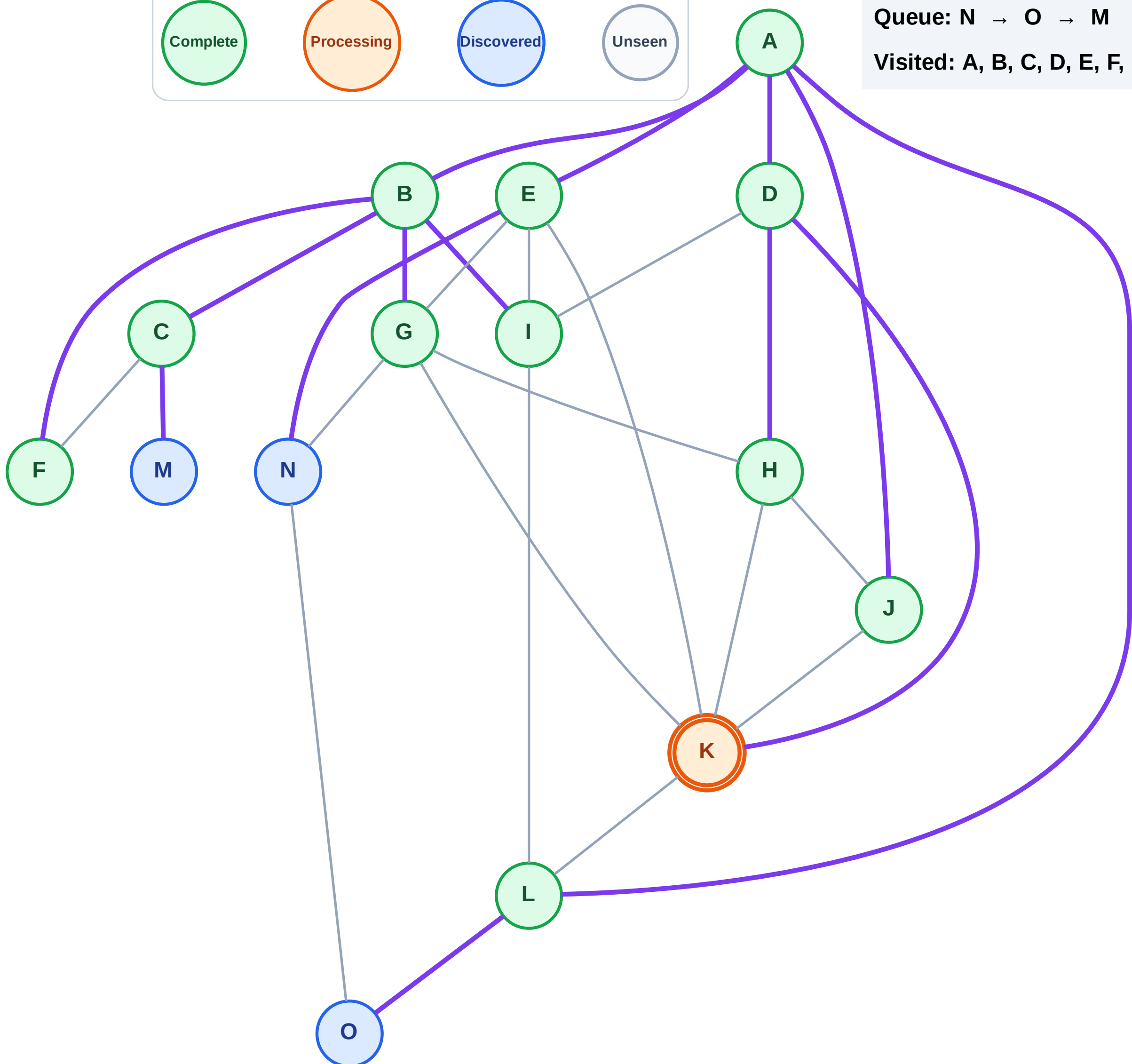
Step 24: Process node K

Legend



Queue: N → O → M

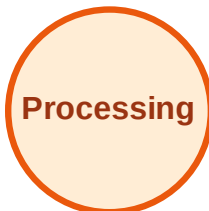
Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

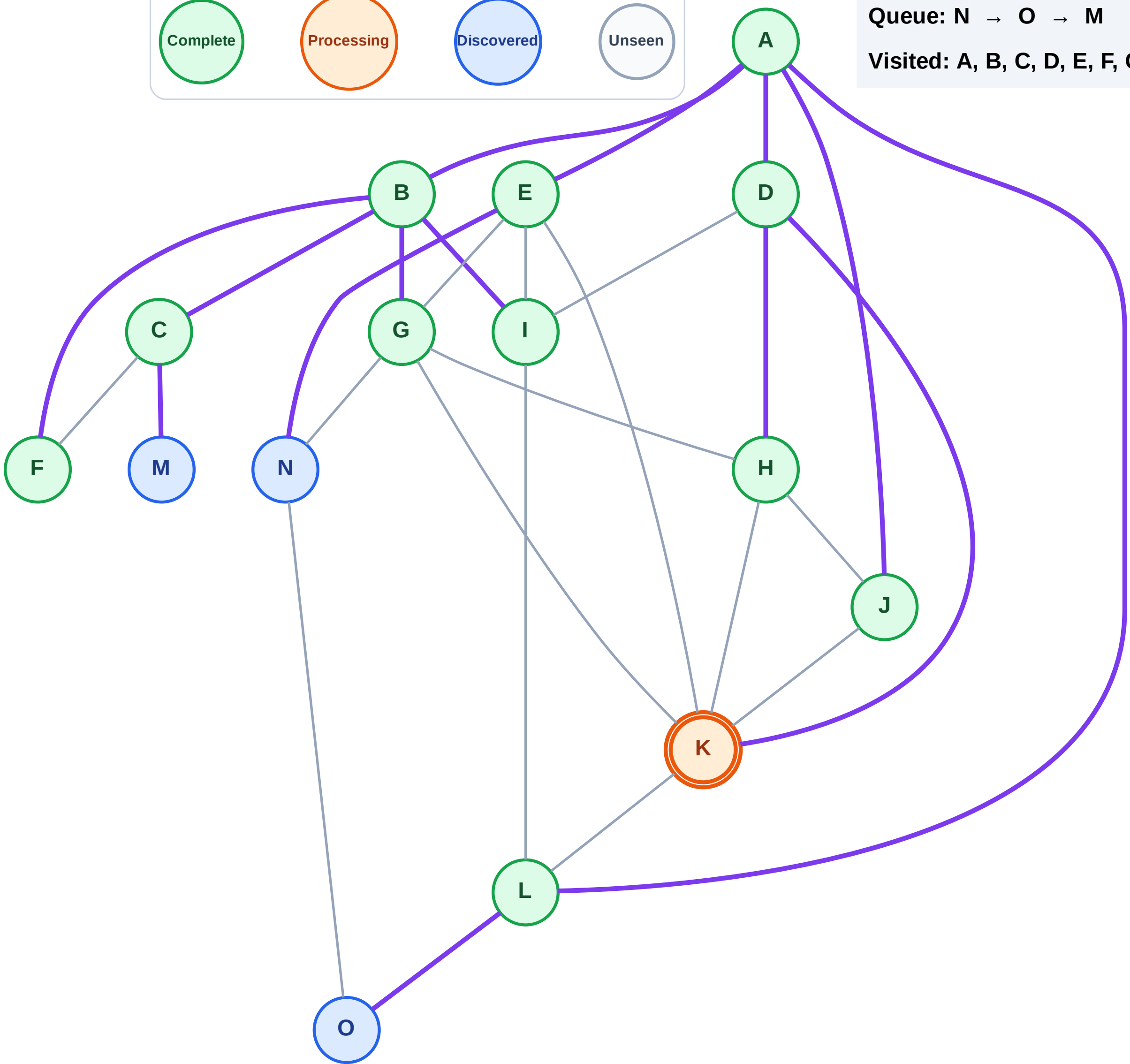
Step 25: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



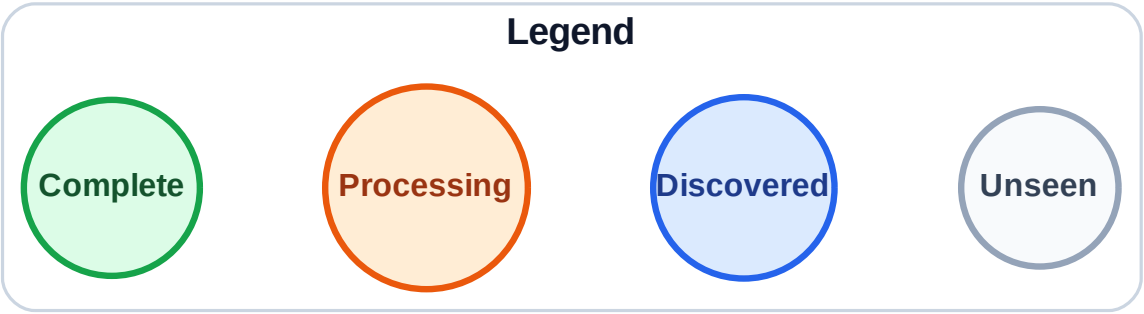
Queue: N → O → M

Visited: A, B, C, D, E, F, G, H, I, J, K, L



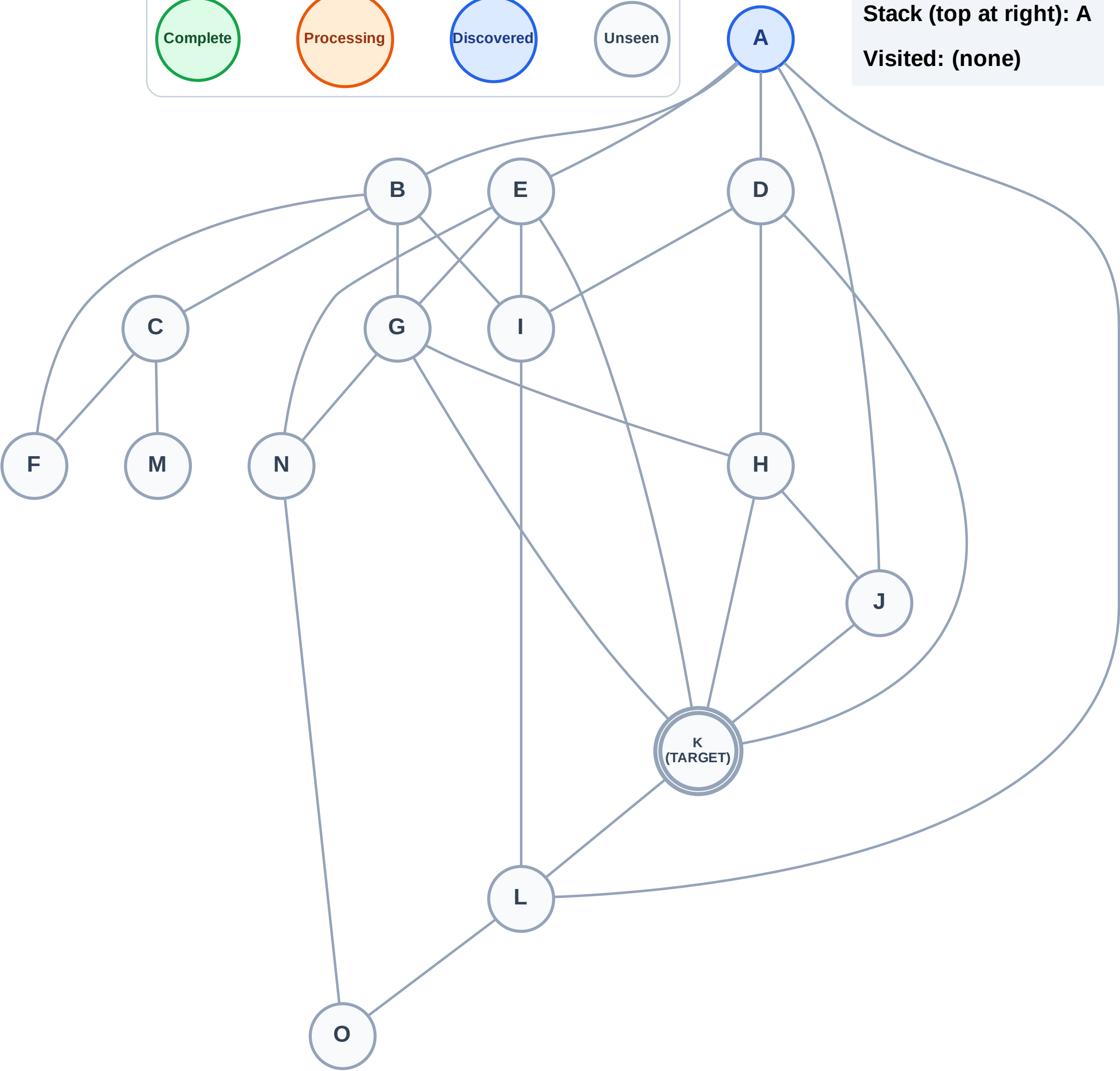
DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

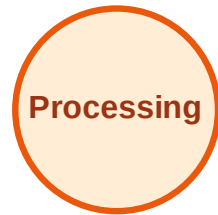
Visited: (none)



DFS Traversal

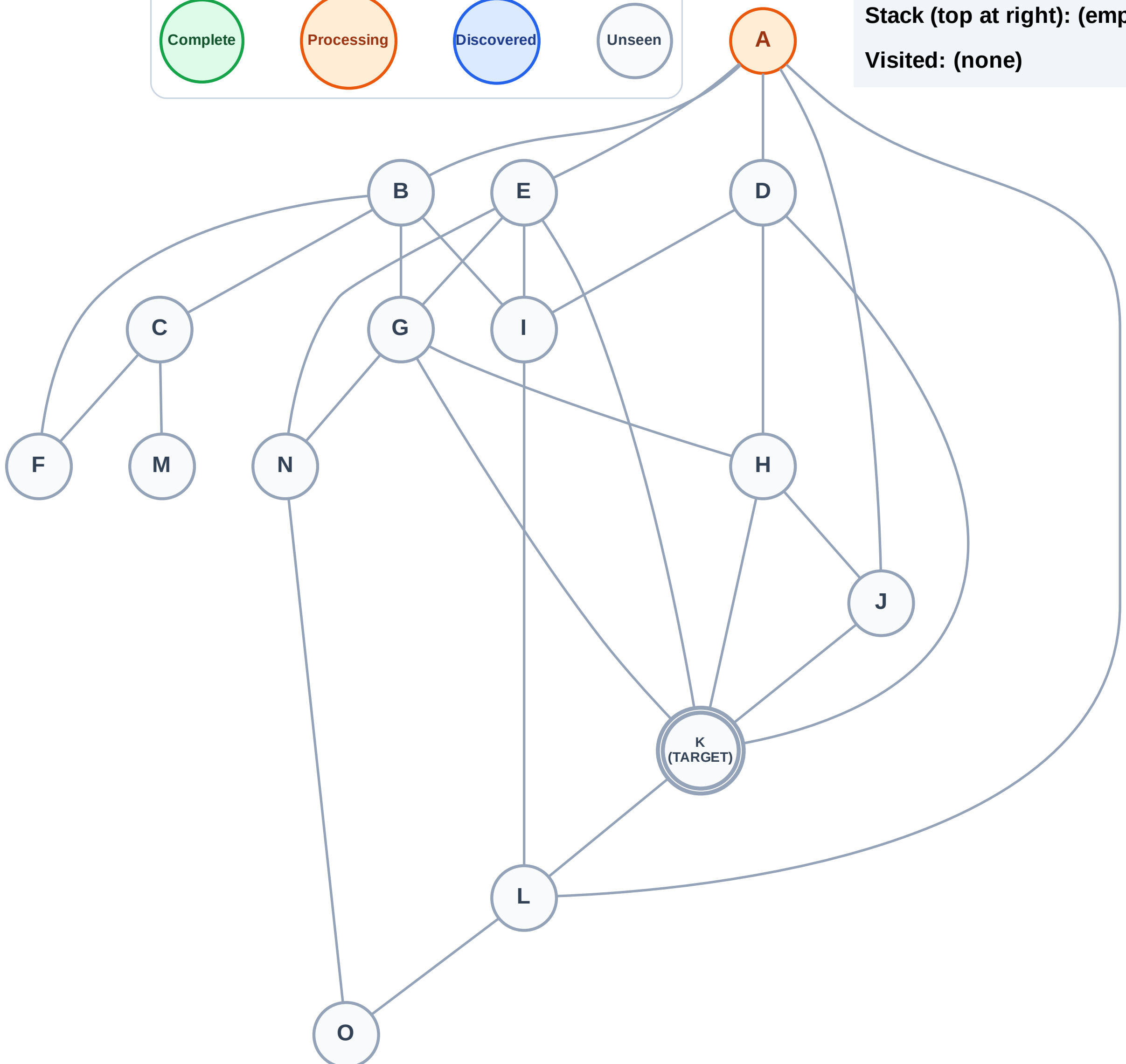
Step 2: Process node A

Legend



Stack (top at right): (empty)

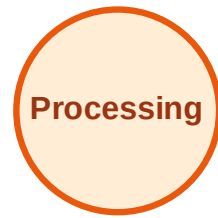
Visited: (none)



DFS Traversal

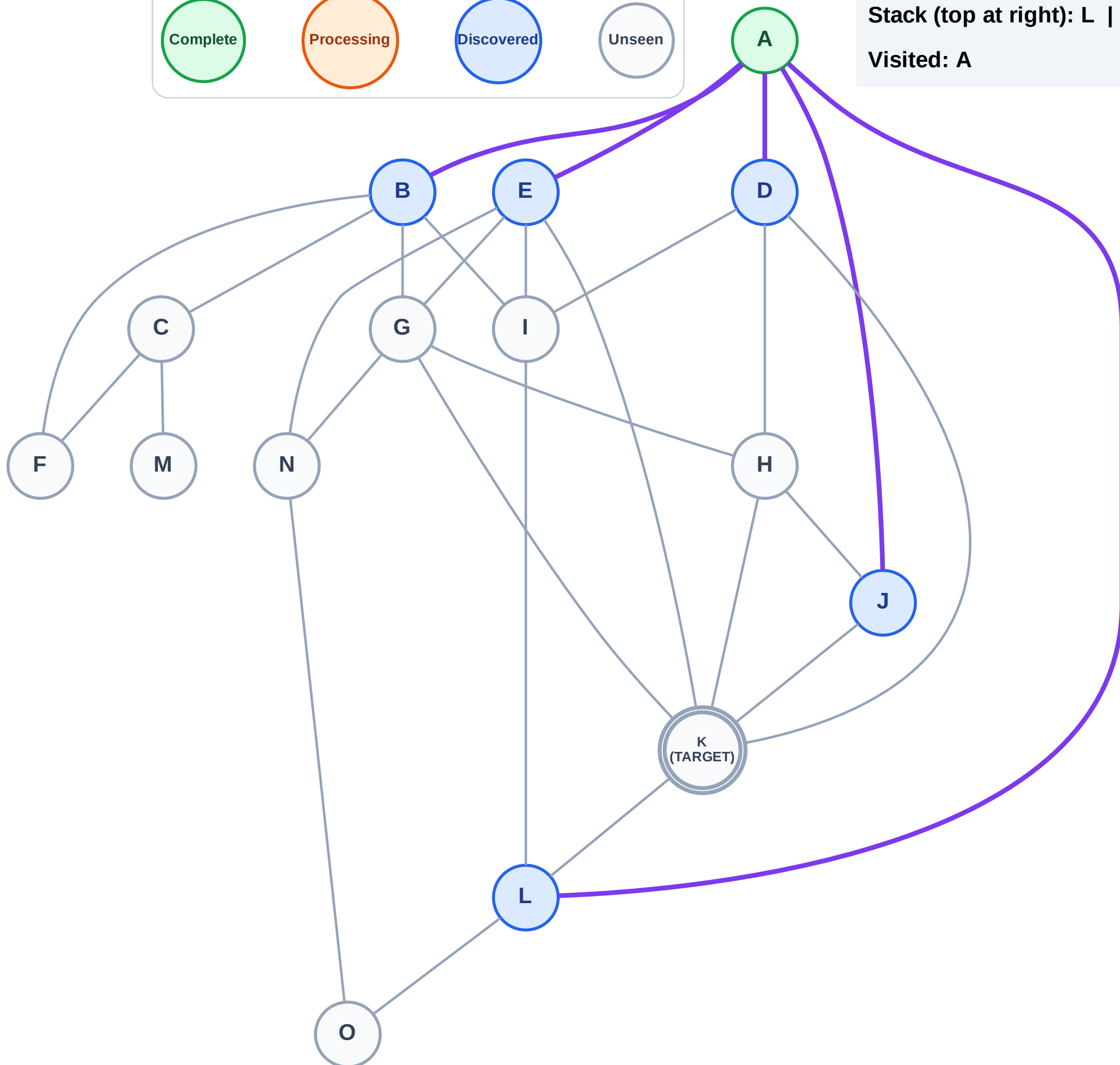
Step 3: Discover L, J, E, D, B from A

Legend



Stack (top at right): L | J | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

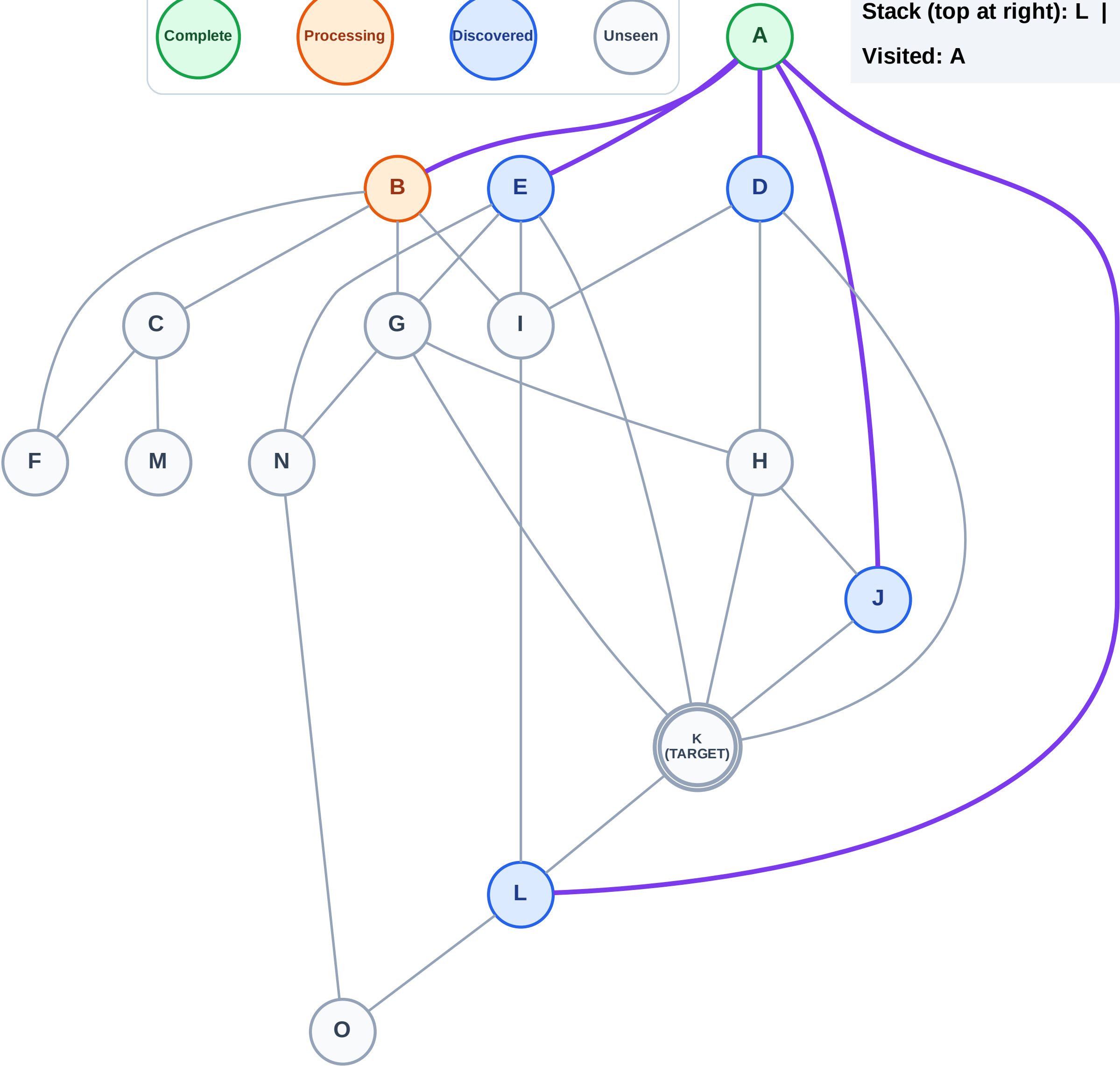
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D

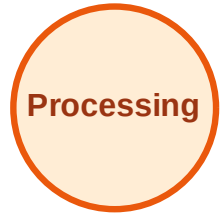
Visited: A



DFS Traversal

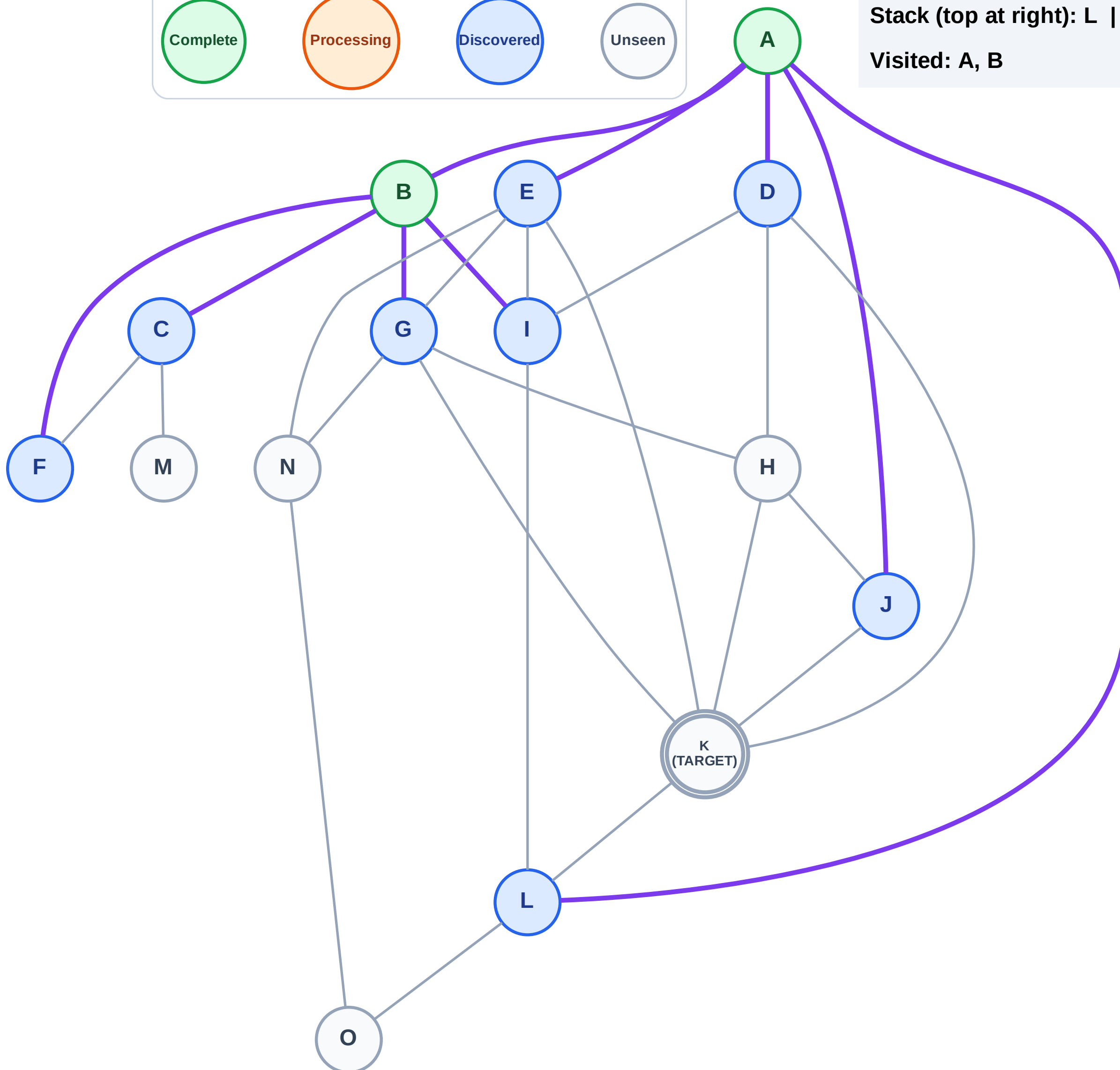
Step 5: Discover I, G, F, C from B

Legend



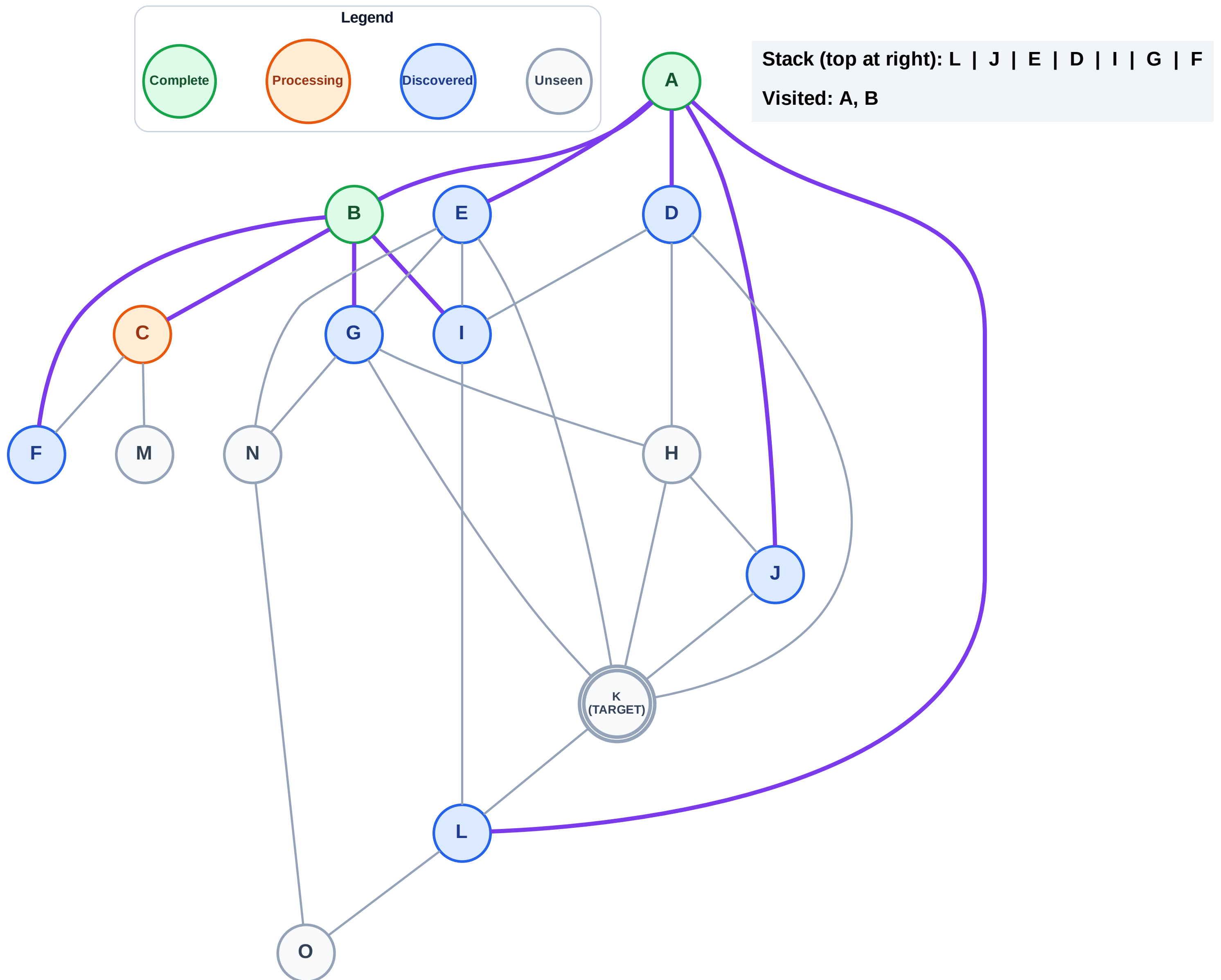
Stack (top at right): L | J | E | D | I | G | F | C

Visited: A, B



Step 6: Process node C

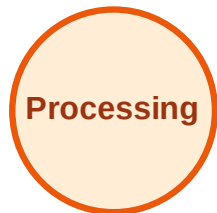
Step 6: Process node C



DFS Traversal

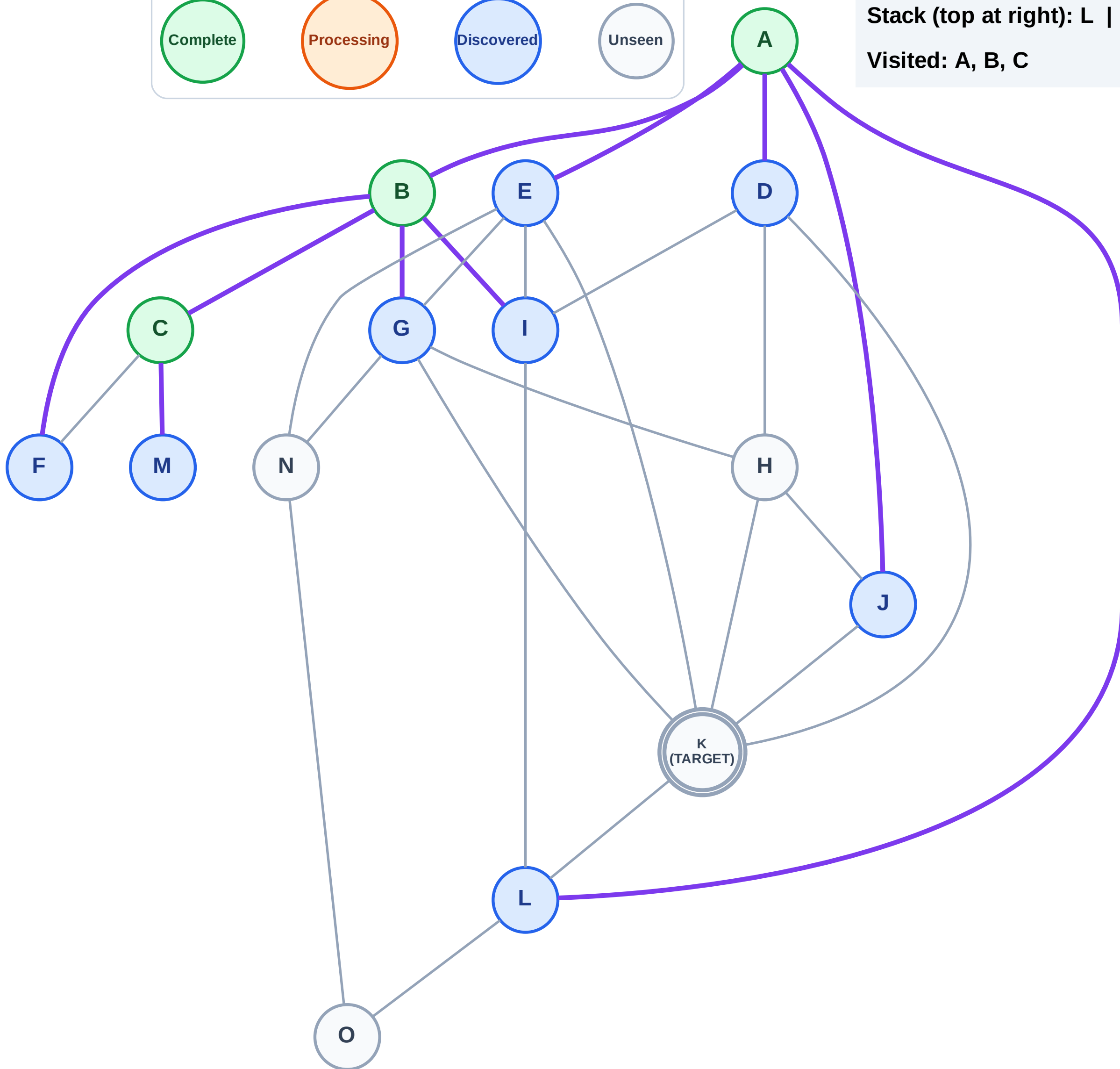
Step 7: Discover M from C

Legend



Stack (top at right): L | J | E | D | I | G | F | M

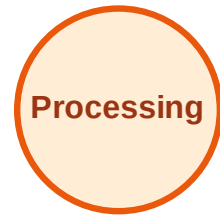
Visited: A, B, C



DFS Traversal

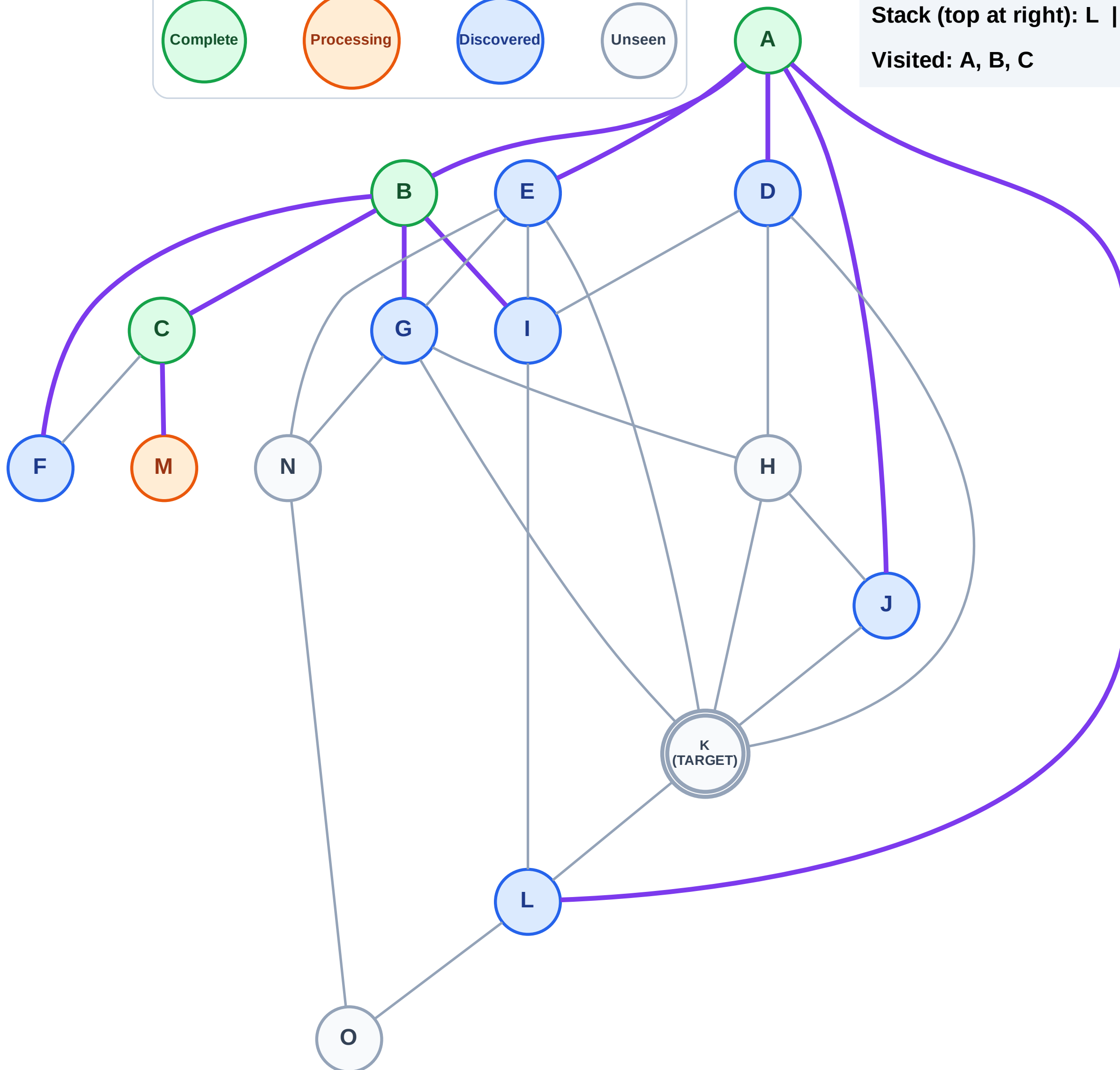
Step 8: Process node M

Legend



Stack (top at right): L | J | E | D | I | G | F

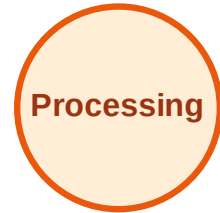
Visited: A, B, C



DFS Traversal

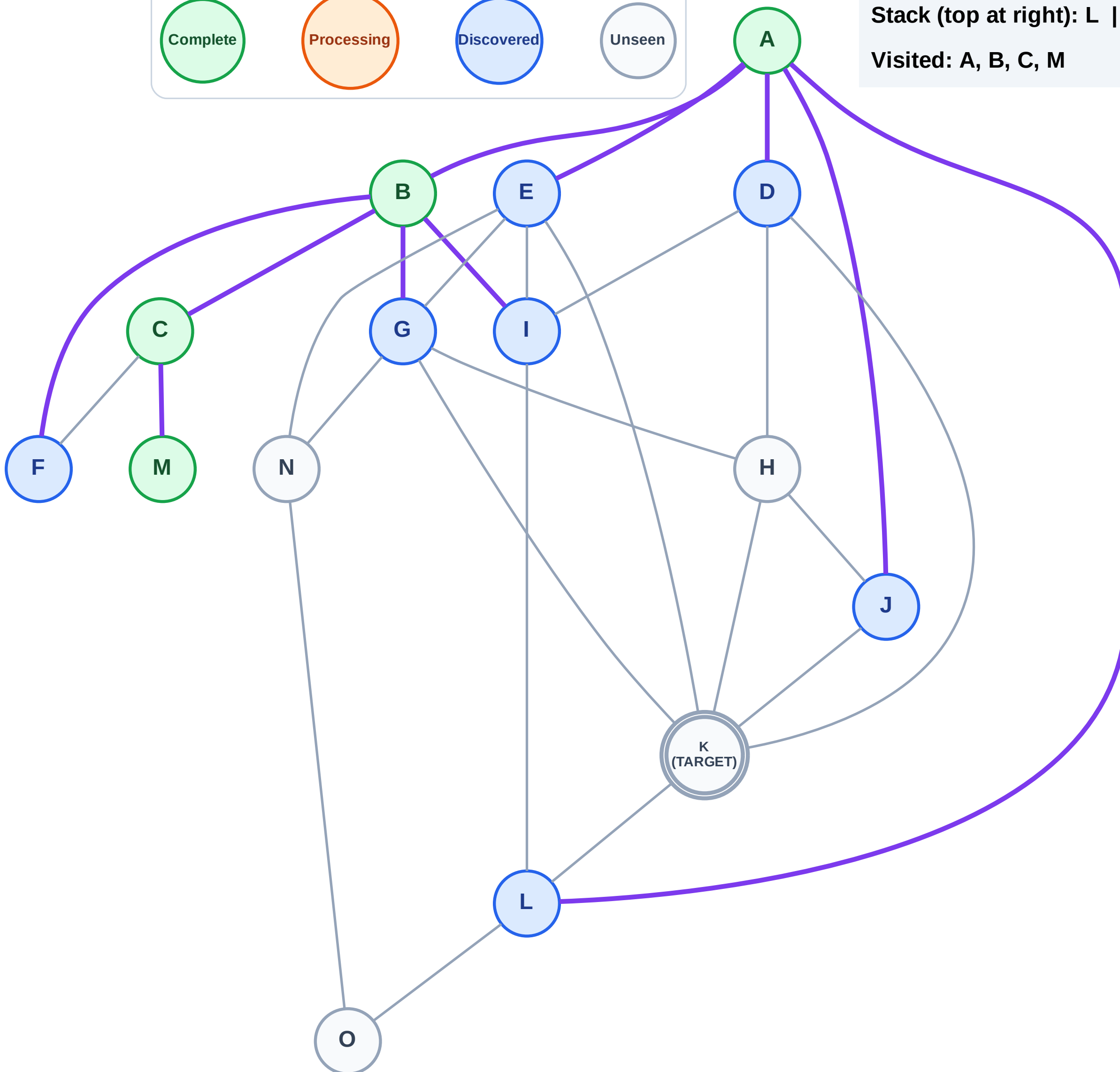
Step 9: No new neighbors from M

Legend



Stack (top at right): L | J | E | D | I | G | F

Visited: A, B, C, M



DFS Traversal

Step 10: Process node F

Legend

Complete

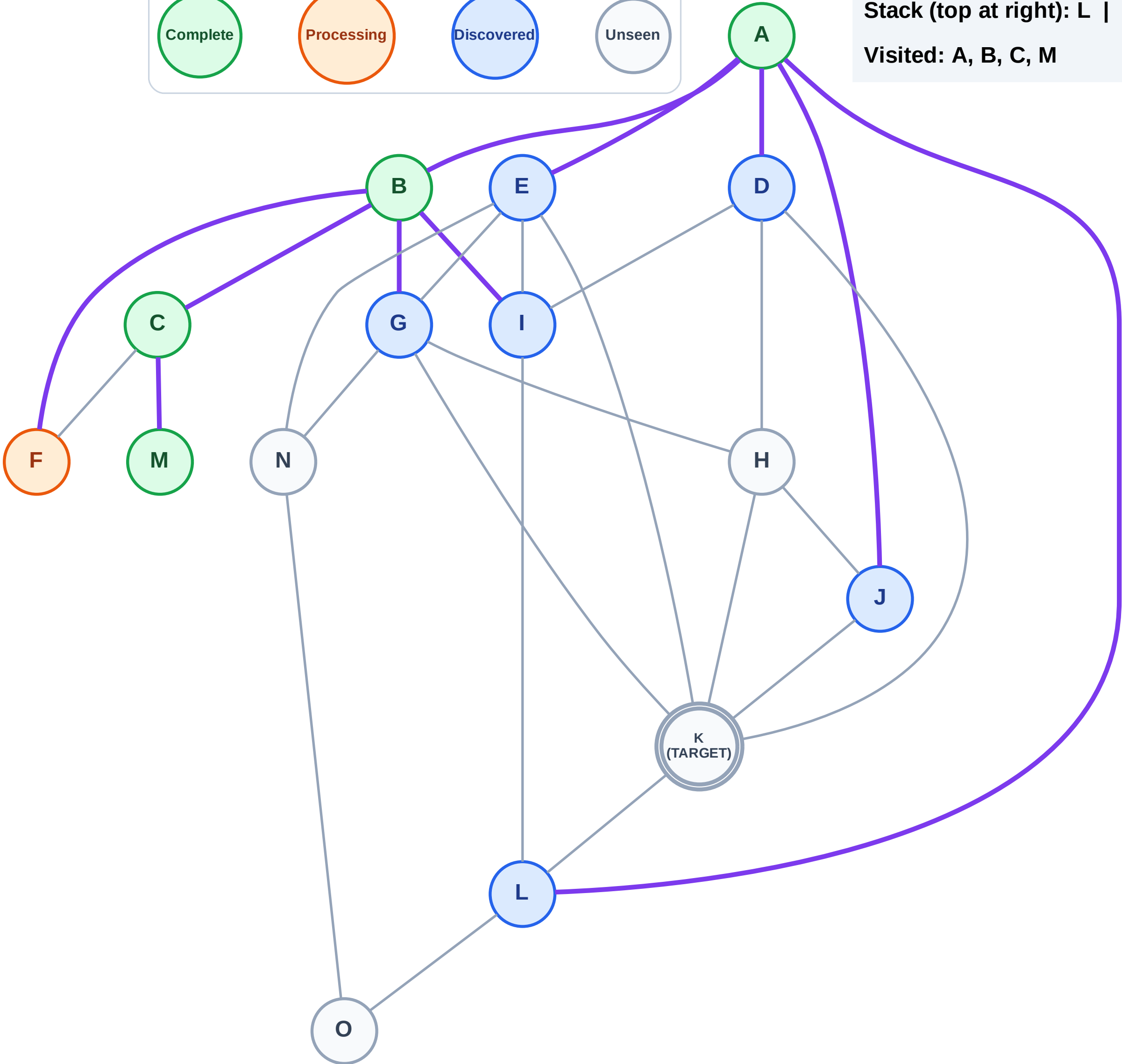
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | G

Visited: A, B, C, M



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

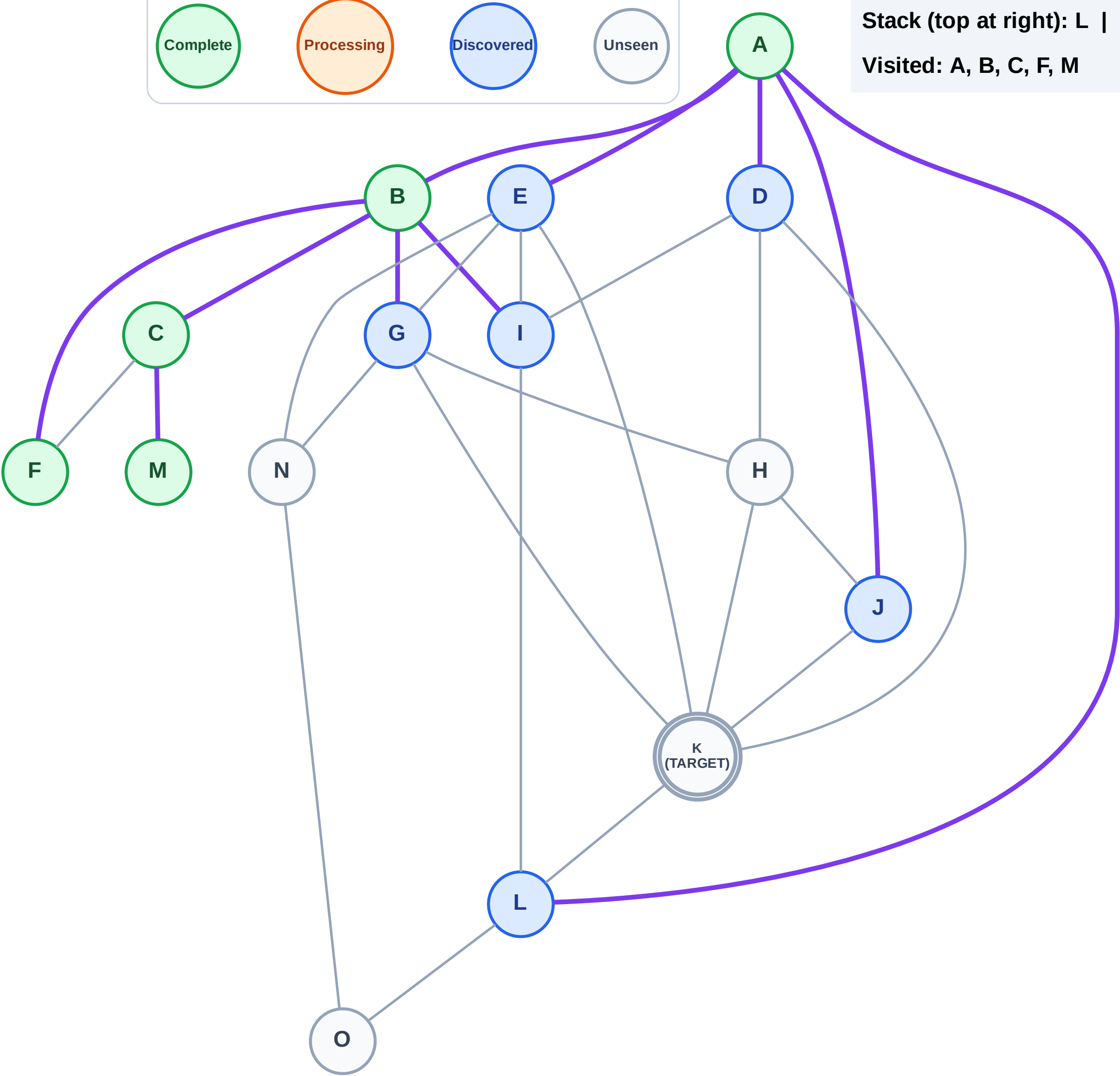
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | G

Visited: A, B, C, F, M



DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



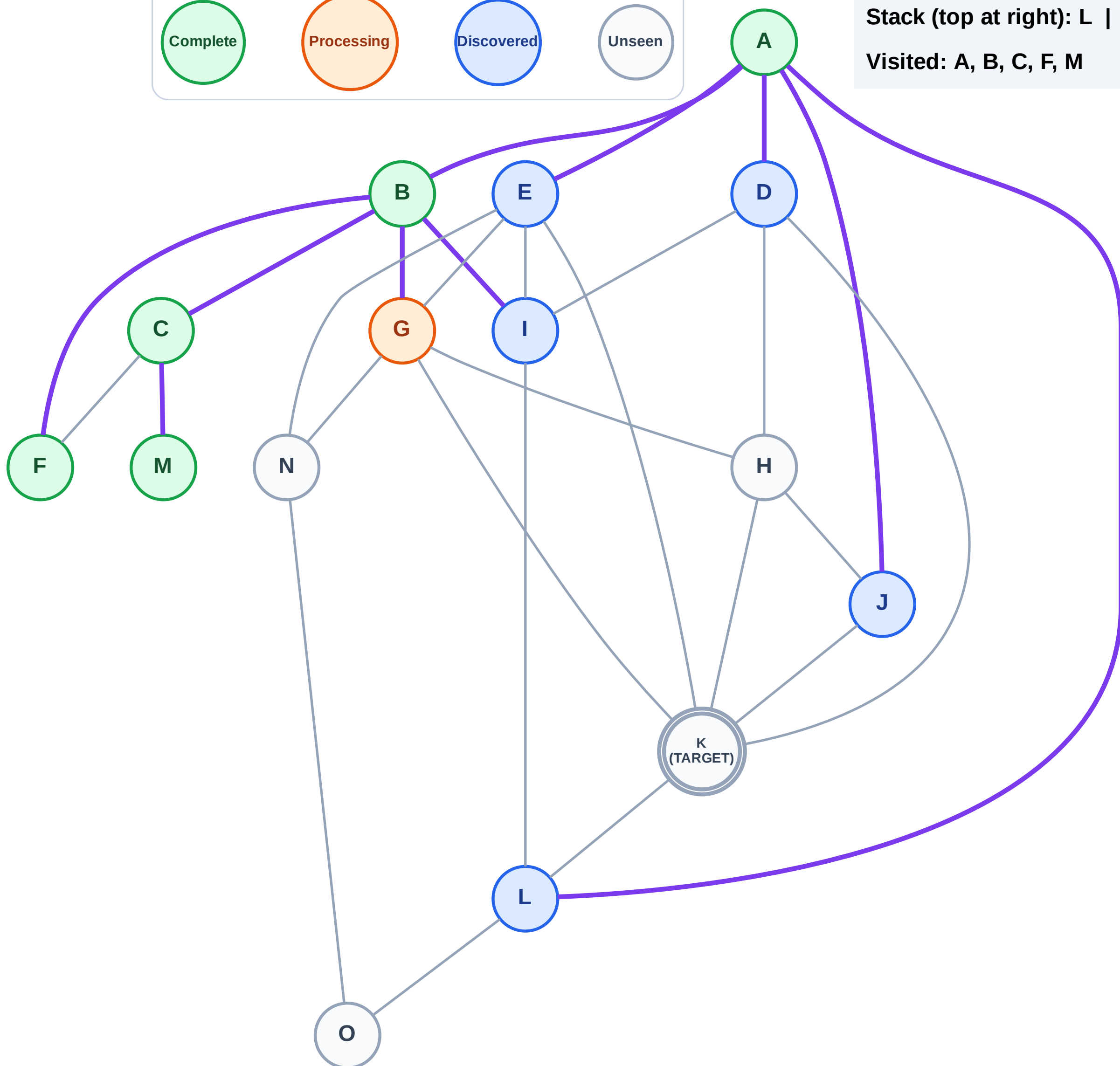
Discovered



Unseen

Stack (top at right): L | J | E | D | I

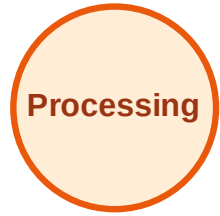
Visited: A, B, C, F, M



DFS Traversal

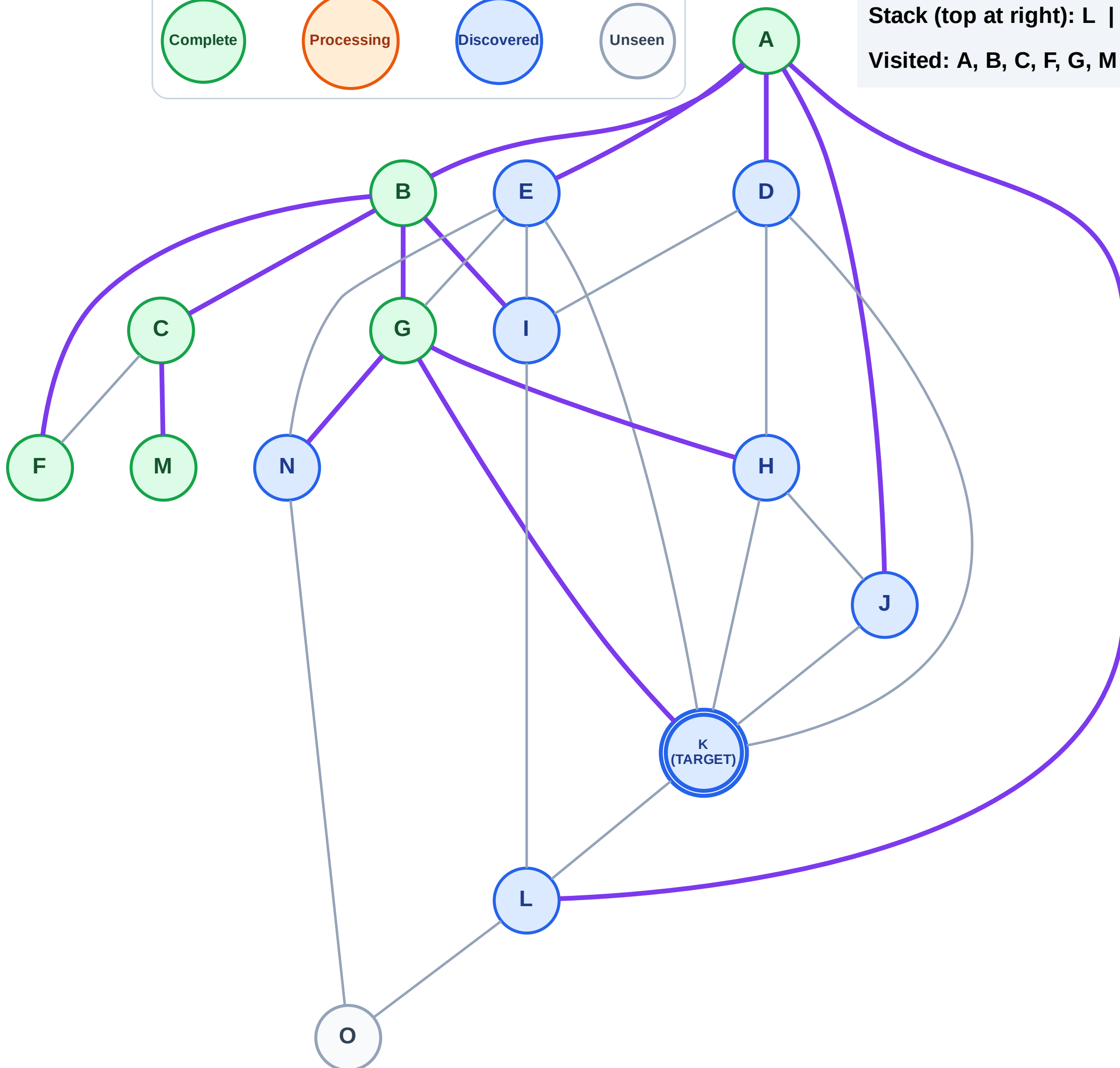
Step 13: Discover N, K, H from G

Legend



Stack (top at right): L | J | E | D | I | N | K | H

Visited: A, B, C, F, G, M



DFS Traversal

Step 14: Process node H

Legend

Complete

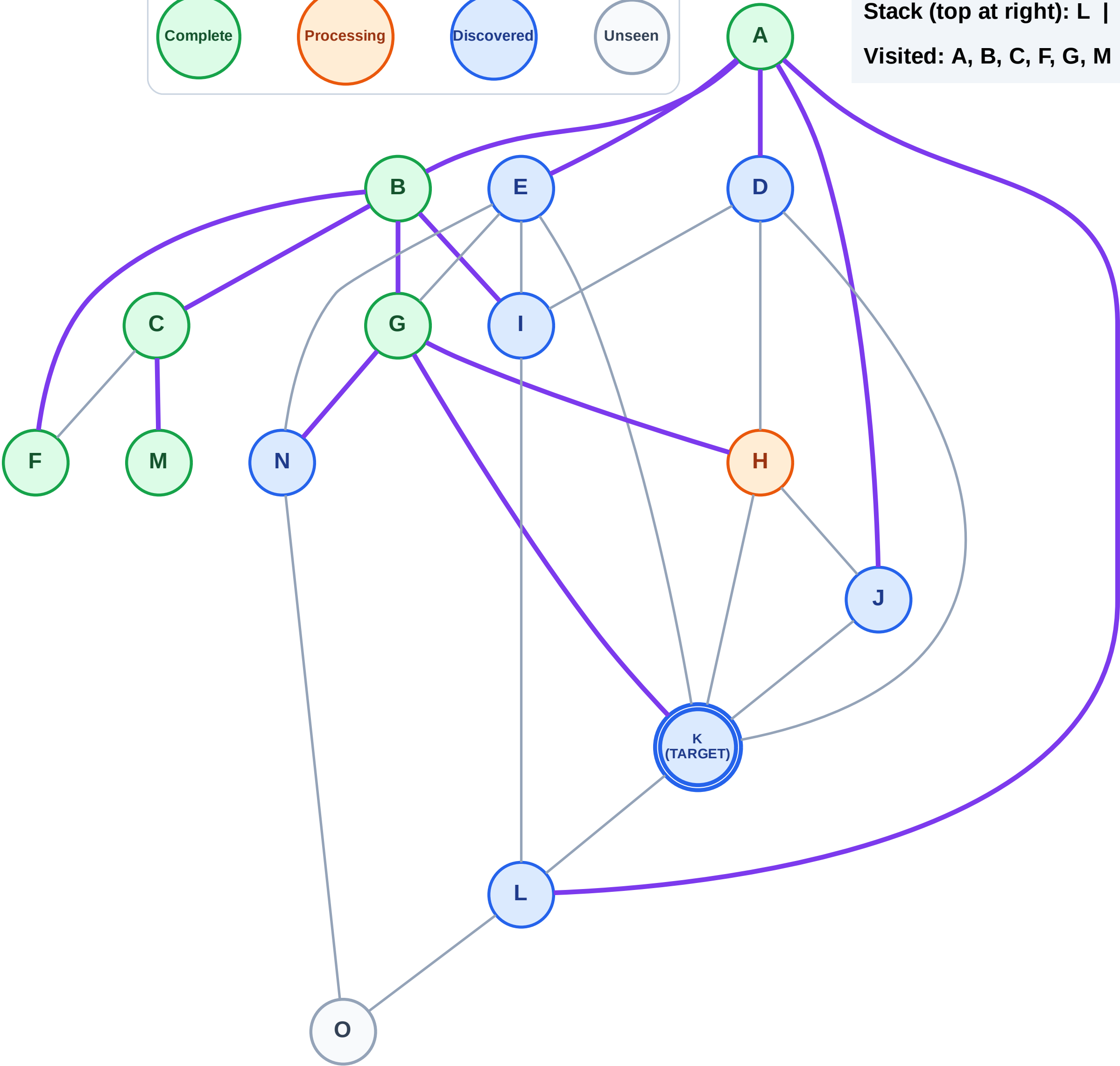
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | N | K

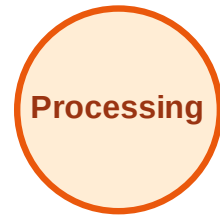
Visited: A, B, C, F, G, M



DFS Traversal

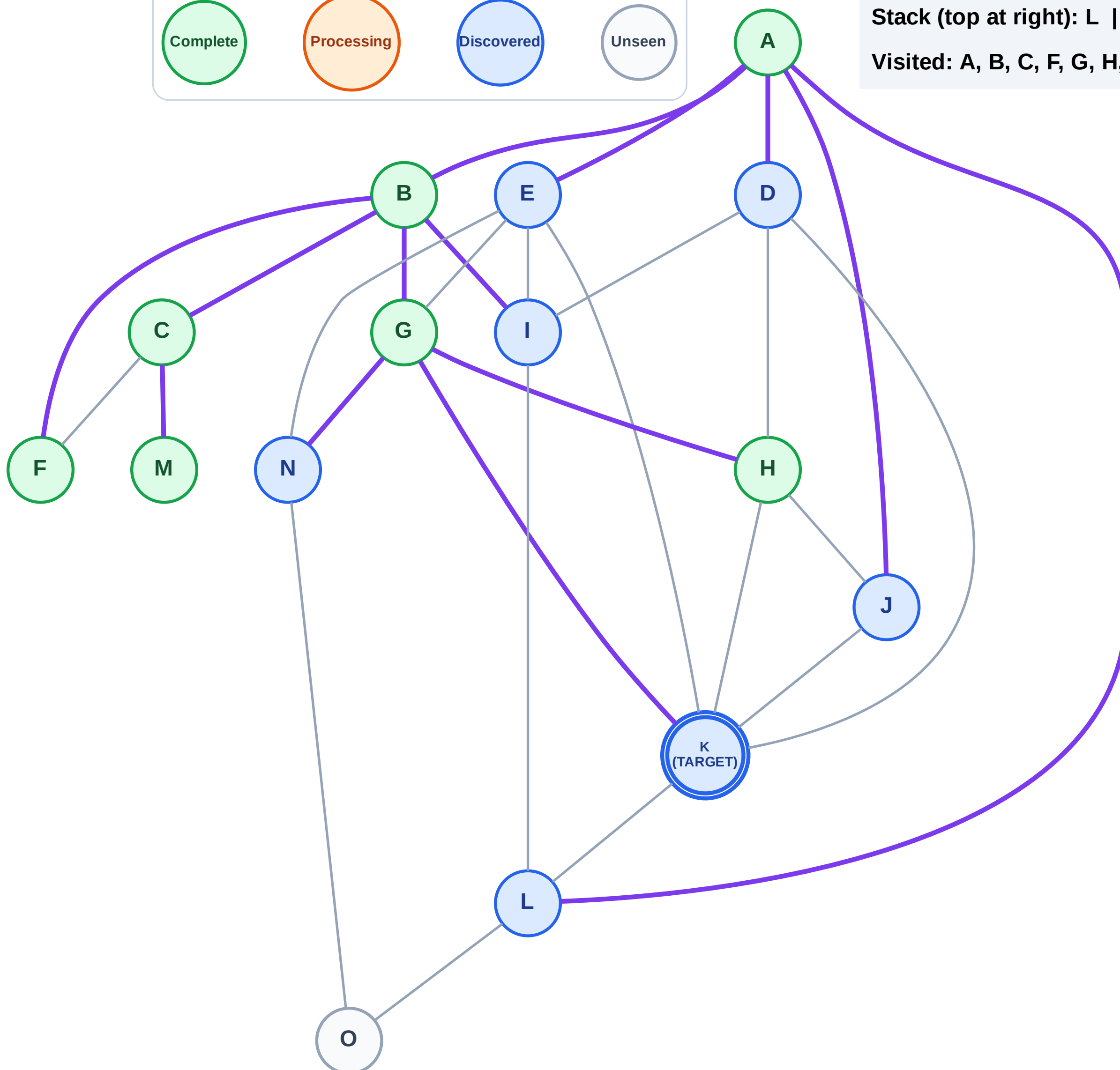
Step 15: No new neighbors from H

Legend



Stack (top at right): L | J | E | D | I | N | K

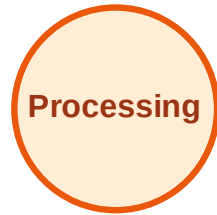
Visited: A, B, C, F, G, H, M



DFS Traversal

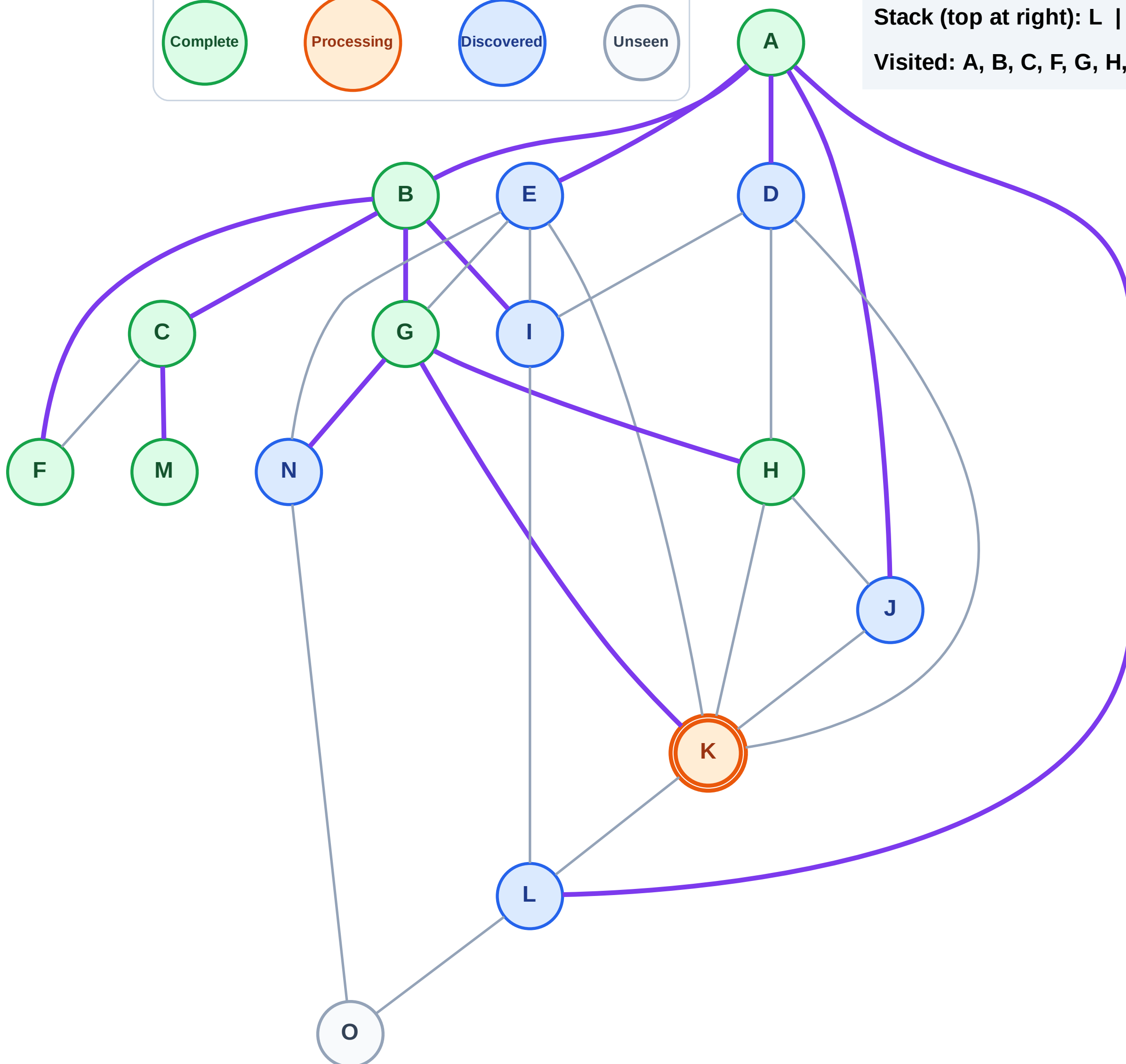
Step 16: Process node K

Legend



Stack (top at right): L | J | E | D | I | N

Visited: A, B, C, F, G, H, M



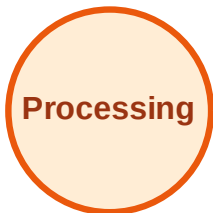
DFS Traversal

Step 17: TARGET NODE K REACHED! Path (3 hops): A → B → G → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): L | J | E | D | I | N

Visited: A, B, C, F, G, H, K, M

